

Reinforcement Learning for Large Language Models: A Comprehensive Survey from Alignment to Agentic Decision-Making

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ABSTRACT

Reinforcement learning (RL) has become a central post-training paradigm for large language models (LLMs), typically deployed alongside supervised fine-tuning, rejection sampling, and distillation, extending their capabilities from instruction following to alignment with human preferences, mathematical and scientific reasoning, autonomous tool use, safety-critical decision-making, and continual self-improvement. Yet the literature that documents this transformation remains fragmented along domain boundaries: alignment surveys omit reasoning and safety; reasoning surveys ignore reward modelling; and agent surveys neglect evaluation and reporting practices. This survey offers a synthesis that combines a cross-domain taxonomy, a quantitative methodological audit, and a reporting checklist for RL for LLMs, a combination that, to the best of our knowledge and per a dated update search through August 2026, no single prior survey provides, organising a curated corpus of 107 representative studies published between 2017 and 2026 into a unified taxonomy of five functional domains: alignment (42 papers), reasoning (22), tool use and agentic behaviour (19), safety and robustness (18), and continual improvement (5), with one pre-LLM foundation. For each domain, we trace the evolution of algorithms, reward designs, and training recipes, from the canonical three-stage RLHF pipeline through direct preference optimisation and its variants to group-relative policy optimisation and RL with verifiable rewards. Our cross-cutting analysis reveals five macro-level trends. First, PPO's early dominance has eroded: measured in the same unit at both ends, PPO-style actor-critic optimisation was used by 9 of the 11 RL-trained systems in the 2020–2022 subset (82%) but by only 35 of the 82 RL-trained systems corpus-wide (42.7%); as a primary algorithm family, PPO accounts for 18.7% of the corpus, with DPO-family methods (14.0%) and GRPO/DAPO (6.5%) absorbing much of the shift. Second, the literature now spans 18 distinct reward signal types, with reward design becoming a domain-specific engineering discipline. Third, evaluation is the weakest methodological link: 76.6% of studies omit safety evaluation entirely, and only three of the 89 non-safety-domain papers include dedicated safety assessment. Fourth, reward hacking (28.0%) and distribution shift (21.5%) are the most frequently reported failure modes, countered by 93 reported mitigation instances that we catalogue across nine strategy categories. Fifth, reporting practices are in crisis: only 5.6% of papers disclose compute budgets and 5.6% report random seeds. Building on these observations, the survey contributes a unified five-domain taxonomy, an 18-type reward signal taxonomy, a decision-support matrix for practitioners, and RL-LLM-REPORT, a 32-item reporting checklist with prevalence baselines derived from the corpus, together with eight open challenges that chart the road ahead for RL-driven language models.

1. Introduction

1.1. Background and Motivation

Large language models (LLMs) have fundamentally reshaped the landscape of artificial intelligence. Systems such as GPT-4 (OpenAI, 2023), LLaMA 2 (Touvron, Martin, Stone, et al., 2023), Claude (Bai, Jones, Ndousse, et al., 2022), and DeepSeek-R1 (DeepSeek-AI, 2025) demonstrate

remarkable capabilities in natural language understanding, code generation, mathematical reasoning, and multi-turn dialogue. These capabilities are achieved through massive-scale pre-training on internet-scale corpora, followed by a critical *post-training* phase that aligns model behaviour with human intentions and safety requirements.

Reinforcement learning (RL) has emerged as a central paradigm for this post-training phase, typically deployed in combination with supervised fine-tuning, rejection sampling, and distillation rather than in isolation. The intellectual lineage of this development reaches back to classical RL (Sutton & Barto, 2018) and to preference-based RL for control (Christiano et al., 2017), but the publication of InstructGPT by Ouyang et al. (2022) marked a decisive inflection point, demonstrating that Reinforcement Learning from Human Feedback (RLHF) using Proximal Policy

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Optimisation (PPO) could transform a pre-trained model into a system that reliably follows instructions, produces helpful responses, and avoids harmful content. This three-stage pipeline, supervised fine-tuning (SFT), reward model training, and RL policy optimisation, rapidly became the industry standard, adopted by Anthropic (Bai, Jones, et al., 2022; Bai, Kadavath, et al., 2022), Google DeepMind (Glaese, McAleese, Trebacz, et al., 2022), and Meta (Touvron et al., 2023).

However, the scope of RL for LLMs has expanded dramatically beyond alignment. In 2023, Rafailov et al. (2023) introduced Direct Preference Optimisation (DPO), which eliminates the reward model entirely by reparameterising the RLHF objective as a simple classification loss, spawning a large family of reward-free variants including SimPO (Meng, Xia, & Chen, 2024), ORPO (J. Hong, Lee, & Thorne, 2024), and KTO (Ethayarajh, Xu, Muennighoff, Jurafsky, & Kiela, 2024). In the reasoning domain, DeepSeek-R1 (DeepSeek-AI, 2025) demonstrated that RL with verifiable rewards can incentivise chain-of-thought reasoning without any human-written demonstrations, while GRPO (Shao et al., 2024) and DAPO (ByteDance Seed and Tsinghua AIR, 2025) showed that the value network can be entirely eliminated. Process reward models (Khalifa et al., 2025; Lightman et al., 2024; Setlur et al., 2024; P. Wang et al., 2024) introduced step-level credit assignment for mathematical reasoning. In tool use, frameworks such as ReAct (Yao et al., 2023), Toolformer (Schick et al., 2023), Reflexion (Shinn, Cassano, Gopinath, Narasimhan, & Yao, 2023), and Voyager (G. Wang et al., 2023) enabled LLMs to interact with external environments, and a rapidly growing family of agentic-RL systems (Cheng et al., 2025; J. Feng et al., 2025; Jin et al., 2025) now trains such behaviour end-to-end. In safety, RL-based red-teaming (Z.-W. Hong et al., 2024; Perez et al., 2022; Samvelyan et al., 2024) and constrained RL (Dai et al., 2023) have become essential tools for probing and hardening model robustness.

This expansion is mirrored in the corpus curated for this survey: of its 107 representative studies, the 2024–2026 period alone accounts for 70.1%, reflecting a field that has grown from a handful of foundational papers in 2017–2022 to a rapidly diversifying research ecosystem spanning five distinct functional domains. A newcomer to the field today confronts dozens of algorithms, an alphabet soup of preference-optimisation objectives, more than fifty evaluation benchmarks, and no shared map of how these pieces relate. This survey provides that map.

1.2. Why Another Survey? Gaps in the Existing Literature

Despite the rapid growth of the primary literature, existing surveys remain fragmented along domain boundaries. Xiao et al. (2025) provided a thorough analysis of DPO variants, but focused exclusively on reward-free alignment methods. K. Zhang et al. (2025) surveyed RL for large reasoning models from a post-DeepSeek-R1 perspective, but did not perform cross-domain synthesis. K. Liu et al. (2025) covered

RL across the LLM lifecycle broadly, but organised the material by lifecycle stage rather than functional purpose and did not analyse evaluation or reporting practices. Chaudhari et al. (2025) provided a rigorous critical analysis of RLHF in ACM Computing Surveys, but their treatment is constrained to the alignment paradigm and predates the GRPO/DAPO wave that reshaped reasoning research in 2024–2025. Most recently, Z. Wang et al. (2026), an evolving arXiv survey whose earlier (2024) versions circulated as an alignment-techniques review before being expanded into a unified policy-gradient treatment of post-training, offered valuable method-type comparisons, but organised the landscape by method type rather than functional domain, and its earlier alignment-focused versions omitted reasoning, tool use, and safety; it also did not examine cross-domain evaluation or reproducibility patterns. We treat all versions of this work as a single prior survey and compare against its latest version.

This fragmentation produces four critical gaps. First, **no unified taxonomy** bridges the five functional domains (alignment, reasoning, tool use, safety, and continual improvement), preventing researchers from identifying cross-domain patterns such as the near-absence of DPO-family methods among reasoning studies or the widespread omission of safety evaluation in non-safety papers. Second, **no cross-domain synthesis** examines algorithmic choices, reward designs, and evaluation practices side by side across all five domains, making it difficult to transfer insights between them. Third, **no analysis of reporting practices** has been conducted across the RL-for-LLM literature; critical gaps, such as the observation that only 5.6% of papers in our corpus report compute budgets or random seeds, have never been quantified. Fourth, **no reporting standard** exists for RL-based LLM training, leaving the community without a shared framework for transparent and reproducible research.

1.3. Scope and Literature Collection

Study design. In design terms, this article is a *narrative survey combined with a structured, purposive evidence map*: it pairs qualitative per-domain synthesis with a quantitative audit of a curated corpus. It is deliberately *not* a systematic review: it does not follow a protocol-driven database search, and, accordingly, we make no population-level prevalence claims: every percentage in this survey is a property of the curated corpus, not an estimate of the field.

Literature collection. The corpus was assembled through expert curation. We manually collected candidate studies from the proceedings of the major machine learning and natural language processing venues (NeurIPS, ICLR, ICML, ACL, EMNLP, NAACL, COLM, TMLR, AISTATS), from ACM Computing Surveys, and from arXiv, complemented by forward and backward citation snowballing from seminal works such as InstructGPT (Ouyang et al., 2022), DPO (Rafailov et al., 2023), and DeepSeek-R1 (DeepSeek-AI, 2025), and by continuous monitoring of the field (venue programmes, arXiv listings, and community trackers). A final dated update search of arXiv listings and venue programmes, targeting both new primary studies and new surveys, was

performed on 8 August 2026; the 2026 surveys it identified are compared in Section 3.3. Duplicate versions of the same work (e.g., a preprint later published at a venue) were consolidated into a single study coded at its latest available version.

Four considerations guided inclusion. (1) *Relevance*: the study must apply RL, or a preference-optimisation objective derived from an RL formulation, to the training, fine-tuning, or evaluation of large language models ($\geq 1\text{B}$ parameters), or constitute an essential conceptual foundation for such work (e.g., preference-based RL (Christiano et al., 2017) or agentic prompting frameworks (Yao et al., 2023) that later RL methods build upon). (2) *Substance*: the study must present empirical results or a substantive methodological contribution rather than a position statement. (3) *Influence*: preference was given to studies that are widely adopted, highly cited, or introduce a technique that subsequent work builds upon. (4) *Coverage*: the corpus was balanced to represent all five functional domains and the full 2017–2026 period, so that cross-domain and temporal comparisons are meaningful. The resulting corpus comprises 107 studies. Two consequences of criteria (3) and (4) deserve emphasis. First, prioritising influential work and deliberately balancing domains makes the sample *purposive*: it is designed to map the methodological landscape, not to estimate field-wide prevalences or publication trends, and all percentages should be interpreted accordingly (this caveat is repeated where the statistics are reported, in Section 9, and in the limitations of Section 12). Second, we make no claim of exhaustiveness, an impossibility in a field producing hundreds of papers per month.

Unit of analysis. The corpus intentionally mixes three strata: (i) empirical studies that train LLM policies (82 studies: 73 assigned to an established named RL or preference-optimisation family and 9 trained with bespoke or framework-specific procedures, itemised as explicit agentic, iterative-RLHF, and pre-LLM subfamilies in Table 8); (ii) contextual foundations that later RL methods build upon, prompting and scaffolding frameworks, supervised or self-supervised tool learning, benchmarks, and environments; and (iii) prior surveys included as comparators. The quantitative tables in Section 9 either report all 107 studies with explicit N/A rows separating the non-training strata, or state a reduced denominator when a statistic applies only to the 82 trained systems (e.g., the algorithm-family analysis of Section 9.3 and the checklist scoring of Section 10.2); the strata, 82 trained systems and 25 non-training/contextual studies, are used consistently throughout Sections 9 and 10, and algorithm, reward, and reproducibility rates should be read with this stratification in mind.

Data extraction and coding. For each study we recorded a consistent set of attributes: publication metadata, base model and scale, RL algorithm and optimisation architecture (a single *primary* label per axis, assigned by principal contribution), reward signal type(s) and feedback source(s)

(multi-label), evaluation benchmarks and paradigm, safety-evaluation coverage (dedicated/partial/none, defined in Section 9.5), self-reported failure modes and mitigations, and the availability of reproducibility artefacts (code, weights, data, hyperparameters, seeds). Extraction was performed by one author; the second author then reviewed every extracted field against the source text (a full-coverage verification pass rather than independent duplicate extraction), flagging discrepancies that were resolved by joint discussion. Because extraction was not independently duplicated, no inter-rater reliability statistic is reported (see the limitations in Section 12); the flagged-and-resolved cases are logged in the codebook’s adjudication log. To make the extraction auditable, we release as supplementary material the complete 107-row study-level evidence table (`evidence_table.csv`), the codebook with attribute definitions, decision rules, and the adjudication log (`codebook.md`), the failure-mode-mitigation instance list (`failure_mitigation_instances.csv`), the RL–LLM–REPORT scoring matrix (`rlllm_report_scoring_matrix.csv`), the deterministic scoring generator (`generate_scoring_matrix.py`), which derives the full item-level scoring matrix from the evidence table alone, and the analysis script (`reproduce_tables.py`); every count and percentage displayed in Sections 9 and 10, including each table cell, each benchmark row, and the full 32-item prevalence column of Table 21, is recomputed from these files and checked by an explicit expected-value assertion in that script. These attributes drive the cross-cutting analysis in Section 9.

1.4. Contributions

This survey makes six principal contributions to the RL-for-LLM literature:

1. **A cross-domain survey spanning all five RL–LLM functional domains.** To the best of our knowledge, unlike prior surveys that focus on alignment (Chaudhari et al., 2025; Z. Wang et al., 2026), DPO variants (Xiao et al., 2025), or reasoning (K. Zhang et al., 2025), this survey synthesises 107 representative studies across alignment, reasoning, tool use, safety, and continual improvement, with dedicated in-depth treatment of each domain (Sections 4–8).
2. **A unified five-domain taxonomy with dual axes.** The proposed taxonomy organises the landscape along functional purpose and optimisation architecture, revealing domain-specific algorithmic and reward-design patterns not visible in single-domain surveys, including the absence of reward-free methods among the corpus’s reasoning studies and the safety evaluation paradox (92% of non-safety papers in the corpus include no safety evaluation).
3. **An 18-type reward signal taxonomy.** To our knowledge, this is the most comprehensive classification of reward mechanisms in the RL-for-LLM literature, extending beyond the binary reward-based/reward-free dichotomy to encompass 18 distinct types including process reward models (Lightman et al., 2024), generative PRMs (Khalifa et al., 2025), classifier-based

red-team rewards (Perez et al., 2022), environment feedback (Yao et al., 2023), and verbal self-reflection (Shinn et al., 2023).

4. **A quantitative cross-cutting analysis of the corpus.** Section 9 charts temporal, algorithmic, reward-design, evaluation, failure-mode, and reporting trends across the 107 studies, providing empirical grounding for the qualitative narrative.
5. **Practitioner guidance.** A decision-support matrix maps each functional domain to recommended algorithms, reward types, KL strategies, evaluation suites, and known pitfalls, and the proposed **RL-LLM-REPORT** 32-item reporting checklist, with prevalence baselines derived from the corpus, offers venues and authors a concrete instrument for improving transparency.
6. **A forward-looking research agenda.** Eight open challenges are distilled from the cross-domain gaps identified in the analysis, ranging from reward-free reasoning to multi-agent RL and longitudinal alignment studies.

1.5. Survey Organisation

The remainder of this survey is organised as follows, and the overall roadmap is visualised in Fig. 1. Section 2 establishes the theoretical foundations: the MDP formulation of text generation, policy-gradient algorithms, the RLHF pipeline, and the reward-free paradigm. Section 3 introduces the unified five-domain taxonomy, traces the historical evolution of the field, and positions this survey relative to prior work. Sections 4–8 provide in-depth treatments of the five functional domains: alignment (Section 4), reasoning (Section 5), tool use and agentic behaviour (Section 6), safety and robustness (Section 7), and continual improvement (Section 8). Section 9 presents the cross-cutting quantitative analysis of the corpus, covering algorithms, reward design, evaluation practices, failure modes, and reporting gaps. Section 10 distils this evidence into practitioner guidance: the decision-support matrix and the RL-LLM-REPORT reporting standard. Section 11 discusses open challenges and future directions, and Section 12 concludes.

2. Foundations of Reinforcement Learning for Language Models

This section establishes the theoretical foundations required to follow the remainder of this survey. Section 2.1 formalises the RL-for-text-generation framework and the principal policy-gradient algorithms. Section 2.2 describes the standard RLHF pipeline and its extensions. Section 2.3 reviews the explicit-reward-model-free preference-optimisation paradigm (commonly abbreviated as “reward-free”) that has reshaped the field since 2023. Section 2.4 introduces the main classes of reward models that recur throughout the domain-specific sections.

2.1. Text Generation as Sequential Decision-Making

The application of reinforcement learning to language model training requires casting autoregressive text generation as a sequential decision-making problem (Sutton & Barto, 2018). Following the formulation introduced by Ziegler et al. (2019) and refined by Ouyang et al. (2022), text generation is modelled as a finite-horizon Markov Decision Process (MDP) defined by the tuple $(S, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$, where the state $s_t = (x, y_{<t})$ consists of the input prompt x concatenated with all tokens generated so far $y_{<t}$; the action $a_t = y_t$ is the next token selected from the vocabulary $\mathcal{V}$; the transition $\mathcal{P}(s_{t+1}|s_t, a_t)$ is deterministic (appending the chosen token to the sequence); the reward $\mathcal{R}$ is typically assigned at the terminal step (sequence-level) or at each step (process-level); and the discount factor γ is commonly set to 1.0 for finite-length sequences.

Under this formulation, the language model policy $\pi_\theta(y_t|x, y_{<t})$ parameterises a distribution over the vocabulary at each step, and the RL objective is to maximise the expected cumulative reward across generated sequences.

2.1.1. Policy-Gradient Algorithms

Three families of policy-gradient algorithms dominate the RL-for-LLM literature. REINFORCE (Williams, 1992) estimates the policy gradient using Monte Carlo returns but suffers from high variance. Advantage Actor-Critic (A2C) reduces variance by subtracting a learned value baseline, as used in early summarisation work (Stiennon et al., 2020). Proximal Policy Optimisation (PPO) (Schulman, Wolski, Dhariwal, Radford, & Klimov, 2017), a first-order successor to trust-region policy optimisation (Schulman, Levine, Abbeel, Jordan, & Moritz, 2015) and the most widely adopted algorithm in RLHF, constrains policy updates via a clipped surrogate objective:

$$\mathcal{L}^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (1)$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the importance sampling ratio, $\hat{A}_t$ is the generalised advantage estimate, and ϵ is the clipping hyperparameter (typically 0.1–0.2). PPO powered InstructGPT (Ouyang et al., 2022), establishing it as the de facto standard for RLHF.

More recently, **Group Relative Policy Optimisation (GRPO)** (Shao et al., 2024) eliminates the value network entirely by computing advantages relative to a group of sampled responses, substantially reducing the training-time memory footprint; the realised saving depends on model sharing, optimiser states, rollout batch size, sequence length, and implementation, rather than being a fixed fraction. GRPO and its variant DAPO (ByteDance Seed and Tsinghua AIR, 2025) have become the dominant algorithms for reasoning-via-RL, powering DeepSeek-R1 (DeepSeek-AI, 2025) and other large-scale reasoning systems; both are examined in depth in Section 5.

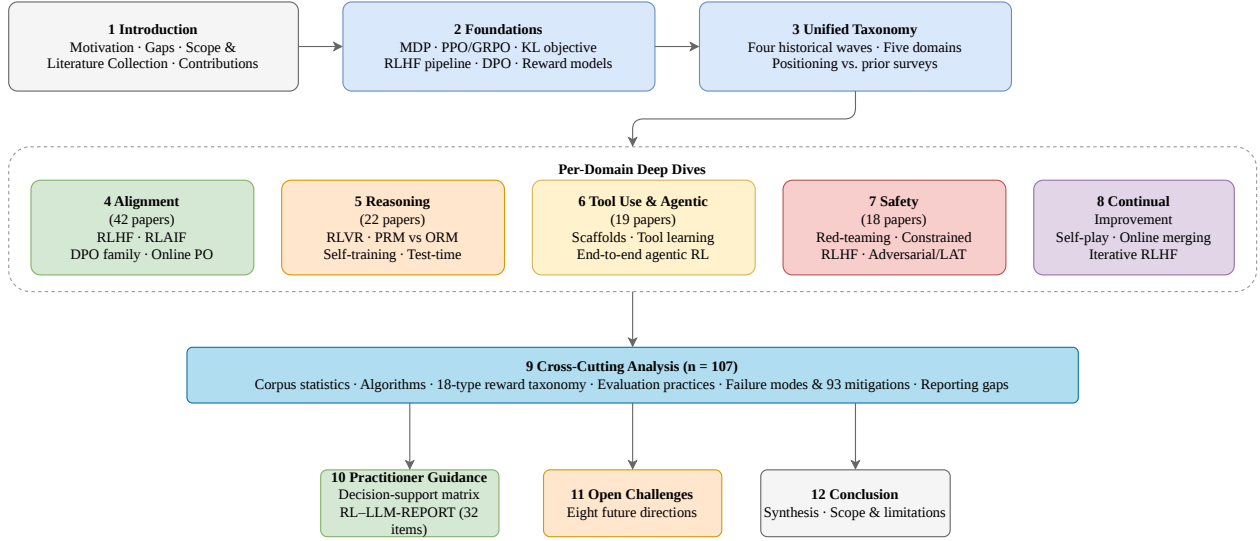


Figure 1: Roadmap of this survey. Foundations (Section 2) and the unified taxonomy (Section 3) feed five per-domain deep dives (Sections 4–8), which are synthesised in the cross-cutting analysis (Section 9) and distilled into practitioner guidance (Section 10) and open challenges (Section 11).

2.1.2. KL-Regularised Optimisation

A central challenge in RL-based LLM training is preventing the policy from deviating excessively from its supervised fine-tuned (SFT) initialisation, which can lead to degenerate outputs, reward hacking, or mode collapse. The standard solution is the KL-regularised objective (Stiennon et al., 2020; Ziegler et al., 2019):

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x)} [r(x, y) - \beta \cdot D_{\text{KL}} [\pi_{\theta}(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x)]] \quad (2)$$

where $r(x, y)$ is the reward model score, π_{ref} is typically the SFT model serving as a reference policy, and $\beta > 0$ controls the strength of the KL penalty. This formulation has a closed-form optimal solution $\pi^*(y|x) \propto \pi_{\text{ref}}(y|x) \exp(r(x, y)/\beta)$, a result that forms the theoretical basis for DPO (Rafailov et al., 2023). The scaling laws for reward model overoptimisation established by L. Gao, Schulman, and Hilton (2023) demonstrated that proxy reward increases monotonically with KL divergence from the reference, while true (gold) reward peaks and then declines, providing empirical evidence for the necessity of KL regularisation.

2.2. The RLHF Pipeline

The standard RLHF pipeline, as established by Christiano et al. (2017) for Atari games and adapted to language models by Ziegler et al. (2019) and Stiennon et al. (2020), consists of three stages, illustrated in Fig. 2.

2.2.1. Stage 1: Supervised Fine-Tuning

A pretrained language model is fine-tuned on a curated dataset of high-quality demonstrations (prompt–response pairs). This stage establishes the initial policy π_{SFT} that serves as both the starting point for RL training and the

reference policy for KL regularisation. Notable SFT datasets include those used by InstructGPT (Ouyang et al., 2022) (human-written demonstrations), Self-Instruct-style corpora (Y. Wang et al., 2023) (self-instruction-generated), and UltraChat (Cui et al., 2024) (multi-turn synthetic dialogue).

2.2.2. Stage 2: Reward Model Training

A separate reward model $r_{\phi}(x, y)$ is trained to predict human preferences from comparison data. The Bradley–Terry (BT) pairwise preference model (Bradley & Terry, 1952) is the standard formulation:

$$p(y_w > y_l | x) = \sigma(r_{\phi}(x, y_w) - r_{\phi}(x, y_l)) \quad (3)$$

where y_w and y_l are the preferred and dispreferred completions, and σ is the logistic function. The reward model is trained via binary cross-entropy loss on human comparison data. This approach was used by InstructGPT (Ouyang et al., 2022), Llama 2 (Touvron et al., 2023), and Sparrow (Glaese et al., 2022). L. Gao, Schulman, and Hilton (2023) established that reward model accuracy scales predictably with model size, providing practical guidance for reward model selection.

2.2.3. Stage 3: RL Policy Optimisation

The SFT model is further trained using PPO (Schulman et al., 2017) to maximise the learned reward while staying close to the reference policy via the KL-regularised objective (Equation 2). This stage requires four models in memory simultaneously: the policy model π_{θ} , the reference model π_{ref} , the reward model r_{ϕ} , and the value model V_{ψ} , making it computationally expensive and memory-intensive. This four-model overhead is a primary motivation for the reward-free alternatives discussed in Section 2.3.

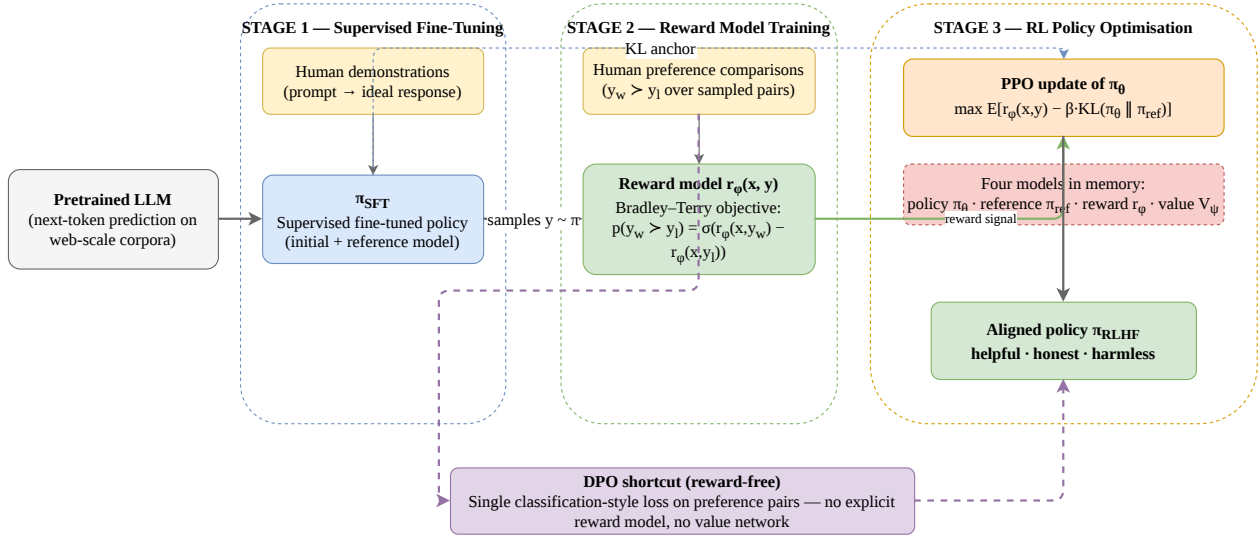


Figure 2: The three-stage RLHF pipeline and its reward-free shortcut. Stage 1 fine-tunes a pretrained LLM on demonstrations to obtain π_{SFT} ; Stage 2 trains a Bradley-Terry reward model r_ϕ from human preference comparisons; Stage 3 optimises the policy with PPO against r_ϕ under a KL constraint to the reference policy. Direct Preference Optimisation (dashed path) collapses Stages 2–3 into a single classification-style objective trained directly on preference pairs, eliminating the explicit reward model and value network.

2.2.4. Extensions of the Pipeline

Several important extensions have been proposed. RLAIIF (Reinforcement Learning from AI Feedback) replaces human annotators with LLM-generated preferences, as demonstrated by Constitutional AI (Bai, Kadavath, et al., 2022) and RLAIIF (H. Lee et al., 2024). Instructable reward models such as SALMON (Sun et al., 2024) allow the reward model to be steered via natural language principles at inference time. Meta-Rewarding (Wu et al., 2024) uses the LLM itself as both generator and judge in an iterative self-improvement loop. Online RLHF (Calandriello et al., 2024; Dong et al., 2024) generates fresh data during training rather than relying on a fixed offline dataset, addressing distribution shift between the training and deployment policies. These extensions collectively expand the RLHF paradigm from a static three-stage pipeline into a dynamic, iterative framework; their alignment-specific instantiations are surveyed in Section 4 and their continual-learning instantiations in Section 8.

2.3. Explicit-Reward-Model-Free Preference Optimisation

The computational overhead and instability of the full RLHF pipeline motivated a family of methods that eliminate the *explicit* reward model entirely, training the policy directly from preference data. A terminological caveat is in order: these methods are conventionally called “reward-free,” and we retain that community shorthand in the tables of Section 9, but the label is imprecise, DPO and its variants still optimise an *implicit* reward under a KL-regularised preference formulation and remain dependent

on preference signals. Throughout this survey, “reward-free” therefore means “without an explicit, separately trained reward model,” not “without a reward.”

Rafailov et al. (2023) demonstrated that the optimal policy under the KL-regularised objective (Equation 2) can be expressed in closed form, enabling a reparameterisation that bypasses reward model training entirely. The DPO loss optimises the policy directly on preference pairs:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right] \quad (4)$$

This eliminates two of the four model roles (reward model and value model) required by PPO-based RLHF, simplifying the training pipeline and reducing memory requirements; as with GRPO, the realised memory saving is implementation-dependent (it varies with parameter sharing, optimiser states, and batch configuration) and should not be read as a fixed 50% reduction. DPO’s simplicity led to rapid adoption, notably in Zephyr (Tunstall et al., 2023) and numerous subsequent alignment systems. The extensive family of DPO variants, IPO, KTO, SimPO, ORPO, CPO, and online extensions, is treated in depth in Section 4.3, and the systematic ablation of this design space by RainbowPO (Zhao et al., 2025) is discussed there. However, DPO-family methods are not without limitations: Park, Rafailov, Ermon, and Finn (2024) showed that DPO can be exploited through length manipulation, and S. Xu et al. (2024) demonstrated that DPO is not universally superior to PPO, with the relative advantage depending on preference data quality and reward landscape complexity.

2.4. Reward Models: Outcome, Process, and Generative

Because reward design recurs as a central theme in every domain of this survey, we briefly introduce the three principal reward-model classes here. **Outcome reward models (ORMs)** score a complete response with a single scalar, as in the BT models of the RLHF pipeline and the solution-level verifiers of Cobbe et al. (2021). **Process reward models (PRMs)** score each intermediate step of a reasoning trace, enabling fine-grained credit assignment; the paradigm was established by Uesato et al. (2022) and Lightman et al. (2024), automated by Math-Shepherd (P. Wang et al., 2024), given a theoretical footing through process advantage verifiers (Setlur et al., 2024), and made data-efficient by generative long-chain-of-thought verifiers such as ThinkPRM (Khalifa et al., 2025). **Generative reward models (Gen-RMs)** (Mahan et al., 2024) frame reward prediction as next-token generation, allowing the verifier itself to exploit chain-of-thought reasoning. The full 18-type reward-signal taxonomy that generalises these classes across all five domains is presented in Section 9.4.

3. A Unified Taxonomy of RL for LLMs

3.1. Historical Evolution

The field's trajectory can be summarised in four waves, visualised in the milestone timeline of Fig. 3. The *foundational wave* (2017–2021) established preference-based RL (Christiano et al., 2017), KL-controlled fine-tuning of language models (Ziegler et al., 2019), RLHF for summarisation (Stiennon et al., 2020), and outcome verifiers for grade-school mathematics (Cobbe et al., 2021). The *alignment wave* (2022–2023) scaled these ideas to instruction-following assistants, InstructGPT (Ouyang et al., 2022), Anthropic's helpful-and-harmless assistant (Bai, Jones, et al., 2022), Sparrow (Glaese et al., 2022), and Llama 2 (Touvron et al., 2023), and introduced AI feedback via Constitutional AI (Bai, Kadavath, et al., 2022). The *simplification wave* (2023–2024) replaced the four-model PPO pipeline with direct preference objectives (Azar et al., 2024; Ethayarajh et al., 2024; J. Hong et al., 2024; Meng et al., 2024; Rafailov et al., 2023), while process supervision matured for reasoning (Lightman et al., 2024; P. Wang et al., 2024). Finally, the *reasoning-and-agents wave* (2024–2026) is defined by RL with verifiable rewards at scale, DeepSeekMath/GRPO (Shao et al., 2024), o1 (OpenAI, 2024), DeepSeek-R1 (DeepSeek-AI, 2025), Kimi k1.5 (Kimi Team, 2025), DAPO (ByteDance Seed and Tsinghua AIR, 2025), and by end-to-end RL training of tool-using agents (Cheng et al., 2025; J. Feng et al., 2025; Jin et al., 2025; X. Li, Zou, & Liu, 2025).

3.2. Five Functional Domains

Based on our analysis of the 107 curated studies, we propose a five-domain taxonomy that organises the RL-for-LLM landscape along a *functional purpose* axis. Each domain is further characterised by its dominant optimisation

architecture (on-policy, off-policy, hybrid, or reward-free) and reward signal type. The taxonomy is depicted in Fig. 4, its quantitative cross-tabulation is analysed in Section 9.2, and each domain receives a dedicated section below.

Domain 1: RL for Alignment (42 papers, 39.3% of the corpus) encompasses all methods that train LLMs to follow human intentions, produce helpful and harmless outputs, and adhere to specified behavioural norms, from the canonical RLHF systems (Bai, Jones, et al., 2022; Ouyang et al., 2022) through Constitutional AI (Bai, Kadavath, et al., 2022) to the DPO family (J. Hong et al., 2024; Meng et al., 2024; Rafailov et al., 2023) and online preference optimisation (Calandriello et al., 2024; Dong et al., 2024). Section 4 provides the full treatment.

Domain 2: RL for Reasoning (22 papers, 20.6%) emerged rapidly following DeepSeek-R1's demonstration (DeepSeek-AI, 2025) that RL can incentivise chain-of-thought reasoning without human-written demonstrations. It is characterised by verifiable reward signals, process reward models (Lightman et al., 2024; P. Wang et al., 2024), value-network-free algorithms (ByteDance Seed and Tsinghua AIR, 2025; Shao et al., 2024), and self-training (A. Singh et al., 2024; Zelikman, Mu, Goodman, & Wu, 2022). Section 5 provides the full treatment.

Domain 3: RL for Tool Use and Agentic Behaviour (19 papers, 17.8%) encompasses methods that enable LLMs to interact with external tools, APIs, and environments, from self-supervised tool learning (Schick et al., 2023) and agentic prompting (Shinn et al., 2023; G. Wang et al., 2023; Yao et al., 2023) to end-to-end RL-trained agents (Cheng et al., 2025; Jin et al., 2025). Its distinguishing feature is the use of environment feedback as the reward signal. Section 6 provides the full treatment.

Domain 4: RL for Safety and Robustness (18 papers, 16.8%) encompasses adversarial training, red-teaming (Z.-W. Hong et al., 2024; Perez et al., 2022; Samvelyan et al., 2024), constrained RL (Dai et al., 2023), rule-based safety rewards (Mu et al., 2024), and latent adversarial training (Sheshadri et al., 2025). It is unique in that every paper in this domain includes dedicated safety evaluation, in stark contrast to the rest of the corpus. Section 7 provides the full treatment.

Domain 5: RL for Continual Improvement (5 papers, 4.7%) is the smallest and most nascent domain, encompassing self-play (Z. Chen, Deng, Yuan, Ji, & Gu, 2024), online model merging (Lu et al., 2024), iterative online RLHF (Dong et al., 2024; Ye et al., 2024), and continual online preference learning (Qi et al., 2024). Section 8 provides the full treatment. (One additional paper, preference-based RL for control (Christiano et al., 2017), is counted within the 107-study corpus but classified as a pre-LLM foundation outside the five functional domains, so the domain counts are $42 + 22 + 19 + 18 + 5 + 1 = 107$.)

3.3. Related Surveys and Positioning

Several surveys have reviewed aspects of the RL-for-LLM landscape, but, to the best of our knowledge (based

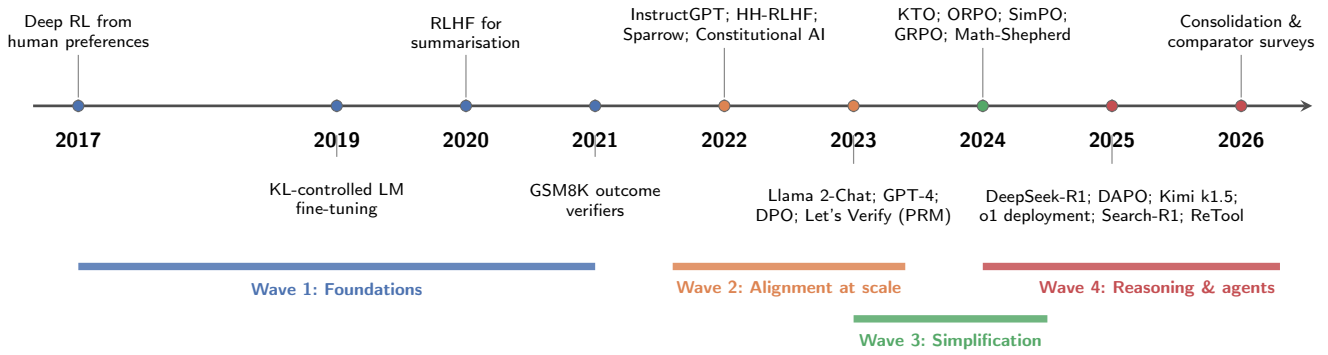


Figure 3: Milestone timeline of RL for LLMs, 2017–2026, drawn on a single horizontal axis with all labels set horizontally for readability. Milestones are organised into four partially overlapping waves: foundations (preference RL, KL-controlled fine-tuning, outcome verifiers), alignment at scale (InstructGPT, HH-RLHF, Constitutional AI, Llama 2), simplification (DPO and its variants, process supervision), and reasoning & agents (GRPO, o1, DeepSeek-R1, DAPO, Search-R1, ReTool). Marker colours indicate the wave of each milestone.

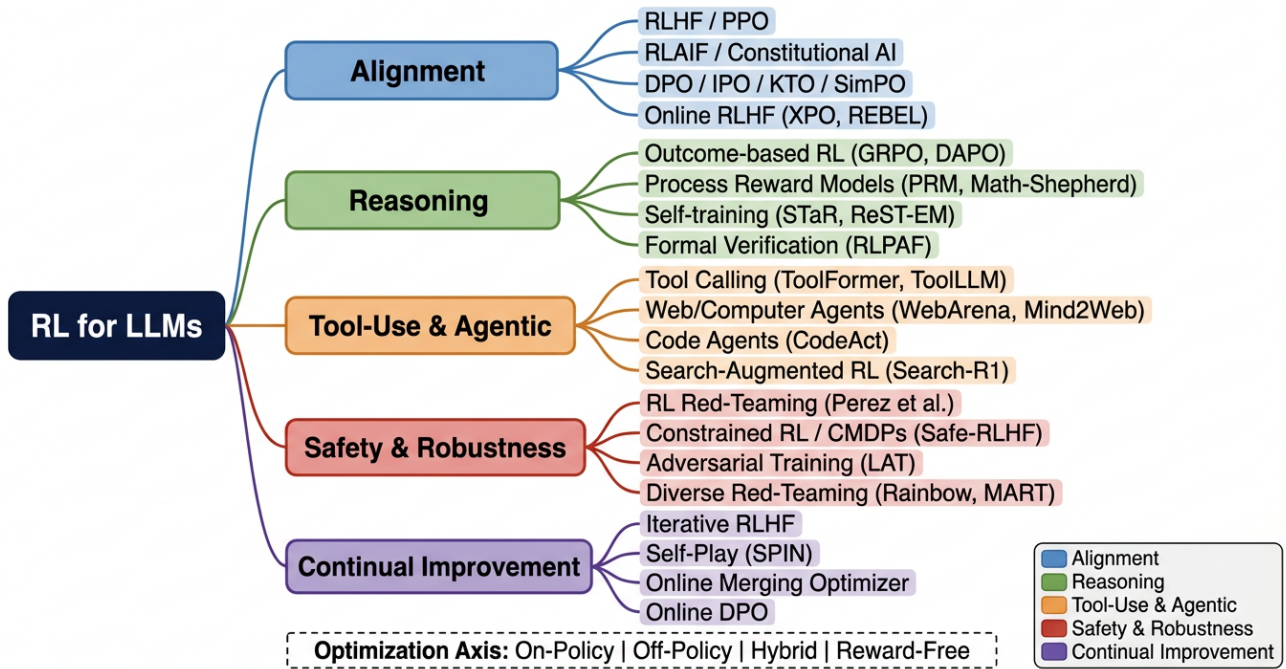


Figure 4: Unified taxonomy of RL for LLMs. The five functional domains (Alignment, Reasoning, Tool-Use & Agentic, Safety & Robustness, and Continual Improvement) are organised along the functional-purpose axis, with representative methods mapped to each domain. The optimisation axis (On-Policy, Off-Policy, Hybrid, Reward-Free) provides the second classification dimension. This taxonomy encompasses all 107 curated studies and 18 distinct reward signal types.

on the survey monitoring described in Section 1.3), none combines the cross-domain taxonomy, quantitative methodological audit, and reporting checklist offered here. Table 1 compares this survey with the five most relevant prior surveys across eight dimensions.

Xiao et al. (2025) provided a focused survey of DPO variants, systematically cataloguing the design space of preference-optimisation objectives; while valuable for understanding the DPO family, it is restricted to reward-free alignment methods. K. Zhang et al. (2025) surveyed reinforcement learning for large reasoning models, offering

a comprehensive post-DeepSeek-R1 perspective on foundational components, core problems, training resources, and applications, but its scope is limited to the reasoning domain. K. Liu et al. (2025) presented a broad survey covering RL applications across the LLM lifecycle from pretraining to deployment, organised by lifecycle stage rather than functional purpose, without cross-domain tabulation. Chaudhari et al. (2025), in their ACM Computing Surveys article “RLHF Deciphered,” provided a critical analysis of RLHF mechanics, the most methodologically rigorous prior treatment, but it is limited to the alignment paradigm and predates the GRPO/DAPO wave and the process-reward-model lineage.

Z. Wang et al. (2026), whose earlier 2024 versions, released under the title “A Comprehensive Survey of LLM Alignment Techniques,” covered RLHF, RLAIIF, PPO, and DPO for alignment only, unified post-training methods in its latest version under a single policy-gradient framework, providing valuable technical comparisons of PPO-, GRPO-, and DPO-based methods, but organised by method type rather than functional domain and, in its alignment-focused versions, omitted reasoning, tool use, and safety. Because all versions share the same arXiv identifier, we compare against this work once, using its latest version. Three 2026 surveys identified by the final update search (Section 1.3) are the closest recent neighbours. C. Zhang (2026) surveys credit assignment in RL for LLMs from reasoning to agentic settings and, like this survey, releases a machine-readable paper inventory and proposes a reporting checklist; its scope, however, is a single methodological axis (credit assignment), and it provides neither a cross-domain functional taxonomy nor a corpus-wide audit of evaluation, safety, and reproducibility practices, so we regard the two checklists as complementary (theirs credit-assignment-specific, ours corpus-wide). X. Wang et al. (2026) surveys reward hacking in large models, deepening one failure mode that our cross-domain failure/mitigation catalogue (Section 9.6) covers in breadth, and Ma et al. (2026) provides a comprehensive treatment of large-model and agent safety that complements our safety-domain synthesis and the safety-evaluation paradox analysis. None of the three combines a cross-domain taxonomy, a quantitative methodological audit, and a corpus-wide reporting standard, so we frame the contribution of this survey as that distinctive combination rather than as a claim of absolute priority. The PRM survey by Zheng et al. (2025) systematically covered the full loop of process-reward-model development, data generation, model training, and downstream usage, but did not position PRMs within the broader RL-for-LLM landscape, and the technical survey of Srivastava and Aggarwal (2025) catalogues algorithms without cross-domain or evaluation analysis.

This survey fills the resulting gap by (1) spanning all five functional domains with dedicated in-depth sections, (2) performing cross-domain quantitative analysis over a curated corpus of 107 studies, revealing patterns invisible to single-domain surveys (e.g., the near-absence of DPO-family methods among reasoning studies and the safety evaluation paradox), (3) compiling the largest reward-signal taxonomy to date, and (4) proposing the RL-LLM-REPORT reporting standard with prevalence baselines.

4. RL for Alignment

Alignment is the largest domain in the corpus (42 papers, 39.3%) and the historical origin of RL for LLMs. This section traces its four principal method families: reward-model-based RLHF systems (Section 4.1), AI-feedback and self-rewarding approaches (Section 4.2), direct preference optimisation and its variants (Section 4.3), and online/iterative preference optimisation (Section 4.4). Section 4.5 reviews

the characteristic pathologies of alignment training, and Table 2 summarises the landscape.

4.1. Reward-Model-Based RLHF Systems

InstructGPT (Ouyang et al., 2022) operationalised the three-stage pipeline of Section 2.2 at scale, showing that a 1.3B-parameter RLHF model could be preferred by human raters over the 175B-parameter GPT-3, an early demonstration that post-training alignment is far more sample- and compute-efficient than pre-training scale for instruction following. Anthropic’s helpful-and-harmless (HH) assistant (Bai, Jones, et al., 2022) extended the recipe with *iterated* online RLHF, retraining the reward model on fresh comparisons collected from the current policy, and documented the helpfulness–harmlessness tension that later motivated constrained formulations (Section 7). Sparrow (Glaese et al., 2022) decomposed the reward into rule-specific violation classifiers plus a preference model and required the agent to support factual claims with evidence, prefiguring the retrieval-grounded alignment of WebGPT (Nakano et al., 2022) and GopherCite (Menick, Trebacz, Mikulik, et al., 2022), which used RLHF (and rejection sampling) to teach models to browse, quote, and cite. Llama 2-Chat (Touvron et al., 2023) demonstrated that the full recipe, separate helpfulness and safety reward models, iterative rejection-sampling fine-tuning, then PPO, could be executed openly and reproducibly at the 7B–70B scale, and Tulu 3 (Lambert et al., 2025) pushed open post-training further, combining SFT, DPO, and a novel RL-with-verifiable-rewards stage while releasing every artefact of the pipeline. Domain-specialised systems such as Qwen2.5-Coder (Hui et al., 2024) adapt the same recipe to code.

On the algorithmic side, Wu et al. (2023) introduced pairwise PPO (P3O), which operates directly on comparative feedback rather than absolute scores, and S. Xu et al. (2024) provided the most careful empirical comparison of PPO against DPO, showing that PPO retains an edge on hard, out-of-distribution, and code-oriented tasks when the reward model is strong, a caution against reading the field’s migration to DPO as a verdict on capability rather than convenience.

4.2. AI Feedback, Constitutional, and Self-Rewarding Approaches

The cost and latency of human annotation motivated a second family that substitutes AI-generated feedback. Constitutional AI (Bai, Kadavath, et al., 2022) trains a harmless assistant without any human harm labels: the model critiques and revises its own outputs against a written constitution (supervised phase), then preference labels for RL are generated by an AI judge (RLAIIF phase). H. Lee et al. (2024) showed that RLAIIF matches RLHF quality on summarisation and dialogue at a fraction of the cost, and introduced direct-RLAIIF, in which an off-the-shelf LLM scores candidates without a distilled reward model. SALMON (Sun et al., 2024) generalised the idea with *instructable* reward models whose

Table 1

Positioning of this survey relative to existing surveys. ✓ = fully addressed; ~ = partially addressed; — = not addressed.

Feature	Xiao et al. (2025)	K. Zhang et al. (2025)	K. Liu et al. (2025)	Chaudhari et al. (2025)	Z. Wang et al. (2026)	C. Zhang (2026)	This survey
All 5 functional domains	—	—	~	—	~	—	✓
Unified dual-axis taxonomy	—	~	—	—	~	~	✓
Reward-signal taxonomy	~	~	—	~	~	—	✓
Cross-domain tabulation	—	—	—	—	—	—	✓
Evaluation-practice analysis	—	~	—	~	~	~	✓
Reporting/reproducibility analysis	—	—	—	—	—	~	✓
Practitioner decision support	~	~	—	—	~	~	✓
Reporting standard proposed	—	—	—	—	—	~	✓

Note: K. Liu et al. (2025) and Z. Wang et al. (2026) receive “~” for domain coverage because they discuss multiple domains but without dedicated per-domain synthesis or cross-domain comparison. The 2024 and 2026 versions of Z. Wang et al. (2026) share the same arXiv identifier (2407.16216) and are treated as a single evolving survey, compared here in its latest version. C. Zhang (2026) receives “~” on taxonomy, evaluation, reporting, and standard because its inventory, gap analysis, and checklist are specific to credit-assignment methods rather than the cross-domain RL-for-LLM literature. Chaudhari et al. (2025) is the most methodologically rigorous prior work but is limited to the alignment paradigm and predates the GRPO/DAPO and PRM developments of 2024–2025.

behaviour can be steered at inference time through natural-language principles, enabling alignment with minimal human supervision. Self-Refine (Madaan et al., 2023) demonstrated the underlying capability, iterative self-feedback and revision, without weight updates, while Meta-Rewarding (Wu et al., 2024) closed the loop by training the model to judge its own judgements, improving both the actor and the judge over successive iterations. Generative reward models (Mahan et al., 2024) unify this direction with reward modelling proper, framing preference prediction as chain-of-thought generation. Collectively these methods have shifted the field’s feedback economy: as quantified in Section 9.4, non-human feedback sources now outnumber human annotation in the corpus by nearly four to one.

4.3. Direct Preference Optimisation and Its Variants

DPO (Rafailov et al., 2023) (Section 2.3) spawned the most active algorithmic family in the alignment domain. IPO (Ψ -PO) (Azar et al., 2024) identified that DPO’s log-sigmoid loss can overfit deterministic preferences and replaced it with a bounded squared objective. KTO (Ethayarajh et al., 2024) eliminated the need for paired preferences entirely, learning from unpaired binary (thumbs up/down) feedback via a prospect-theoretic utility, which broadens the applicable data regime to production feedback logs. SimPO (Meng et al., 2024) removed the reference model, using length-normalised sequence log-probability as the implicit reward with a target margin, simultaneously simplifying training and mitigating length bias. ORPO (J. Hong et al., 2024) folded preference optimisation into the SFT loss through an odds-ratio penalty, collapsing two training

stages into one. CPO (H. Xu et al., 2024) added sequence-likelihood calibration with an SFT anchor for machine translation. sdPO (Kim et al., 2024) partitioned the preference data into curriculum-style stages with progressively updated reference models, and the pairwise cringe loss (J. Xu, Lee, Sukhbaatar, & Weston, 2024) generalised binary-feedback training to iterative preference optimisation. Zephyr (Tunstall et al., 2023) established the practical recipe, distilled SFT on UltraChat followed by distilled DPO on UltraFeedback (Cui et al., 2024), that made DPO the default open-source alignment method in 2024.

Two consolidating works bring order to this family. RainbowPO (Zhao et al., 2025) decomposed the design space into orthogonal components (length normalisation, reference policy, margin, mixing) and showed that no single published variant dominates; a tuned combination outperforms all of them. POWER (Rashidinejad & Tian, 2024) addressed the statistical pathologies of offline preference optimisation, spurious length exploitation and preference collapse, through weighted-entropy robust rewards with dynamic labels. On the theory-versus-practice frontier, Park et al. (2024) disentangled length from quality in DPO’s implicit reward and Azar et al. (2024) formalised the overfitting failure mode; both analyses are corroborated by the failure-mode statistics of Section 9.6.

4.4. Online and Iterative Preference Optimisation

Offline DPO inherits a distribution-shift liability: the policy is optimised on preferences over responses it no longer generates. A third family therefore re-introduces on-policy data collection while retaining reward-free objectives. Iterative online RLHF (Dong et al., 2024; Ye et al.,

2024) alternates between sampling from the current policy, labelling with a (general) preference model, and DPO-style updates, and Calandriello et al. (2024) unified this class theoretically, showing that online IPO with self-play converges toward the Nash equilibrium of the preference game. XPO (Xie et al., 2024) augments online DPO with an optimism-based exploration bonus and provides sample-efficiency guarantees; REBEL (Z. Gao et al., 2024) reduces policy optimisation to iterative regression of relative rewards, recovering PPO-level performance without clipping heuristics or value networks; and OFS-DPO / LFS-DPO (Qi et al., 2024) tackle catastrophic forgetting in continual online preference learning through fast-slow LoRA module competition, foreshadowing the continual-improvement domain of Section 8. Multilingual RLHF (Dang et al., 2024) shows that online preference optimisation transfers across languages with strong cross-lingual generalisation.

4.5. Pathologies of Alignment Training

Three recurring pathologies deserve emphasis. *Reward overoptimisation*: L. Gao, Schulman, and Hilton (2023) established scaling laws showing gold reward peaks and then declines as the policy diverges from the reference, the canonical quantification of reward hacking. *Length and style exploitation*: DPO-family objectives are particularly susceptible to verbosity bias (Meng et al., 2024; Park et al., 2024). *Alignment tax*: preference alignment can degrade base capabilities; Lin et al. (2024) characterised this tax and showed that simple model-averaging between the SFT and RLHF policies recovers much of the Pareto front, a finding operationalised online by Lu et al. (2024). These pathologies, their prevalence, and the full mitigation catalogue are analysed quantitatively in Section 9.6.

5. RL for Reasoning

Reasoning is the second-largest and fastest-growing domain (22 papers, 20.6%), transformed in 2024–2025 by the demonstration that large-scale RL with verifiable rewards (RLVR) can elicit long-horizon chain-of-thought behaviour. We organise the domain into RLVR at scale (Section 5.1), process versus outcome supervision (Section 5.2), self-training (Section 5.3), test-time scaling (Section 5.4), and formal theorem proving (Section 5.5).

5.1. RL with Verifiable Rewards at Scale

The defining property of the reasoning domain is that correctness is often *checkable*: a mathematical answer matches or it does not, code passes tests or it fails. This converts the reward-modelling problem of alignment into deterministic verification, removing reward-hacking pressure on a learned proxy. DeepSeekMath (Shao et al., 2024) introduced GRPO, computing token advantages from group-normalised outcome rewards over G sampled solutions per prompt and thereby deleting the value network; the same recipe, scaled up, produced DeepSeek-R1 (DeepSeek-AI, 2025), whose R1-Zero variant showed that pure RL on a base model, no SFT stage, spontaneously develops reflection, verification,

and “aha-moment” behaviours, at the cost of readability issues that the full R1 pipeline repairs with a small cold-start SFT phase. OpenAI’s o1 (OpenAI, 2024) independently established the paradigm of large-scale RL that lengthens deliberation, and Kimi k1.5 (Kimi Team, 2025) extended RLVR to multimodal inputs with long-context (128K) RL and length-penalty control, contributing the long2short distillation of long-CoT competence into short-CoT budgets.

DAPO (ByteDance Seed and Tsinghua AIR, 2025) open-sourced the systems knowledge behind this wave, identifying four fixes that make large-scale GRPO stable: decoupled (asymmetric) clipping to preserve exploration, dynamic sampling that discards zero-gradient groups whose responses are all correct or all wrong, token-level (rather than sample-level) loss aggregation for long chains, and overlong-response shaping. Qwen2.5-Math (Yang et al., 2024) embedded GRPO within a self-improvement loop in which the model’s own best-of- N outputs, filtered by a reward model, iteratively enrich the training data, while WizardMath (Luo et al., 2025) showed that reinforcement from evolved instructions (RLEIF), combining an instruction-quality RM with an answer-supervised PRM, lifts sub-10B models to competitive mathematical performance. Z. Zhang et al. (2025) distilled cross-run lessons for RLVR practice, including the sensitivity of results to verifier strictness and data curation. Notably, and as quantified in Section 9.2, *no DPO-family method appears among the reasoning studies in our corpus*. This is a property of the curated corpus rather than of the literature as a whole: step-wise preference-optimisation methods such as Step-DPO (Lai et al., 2024) and Full-Step-DPO (H. Xu et al., 2025) have begun to adapt offline preference objectives to long-chain mathematical reasoning. Nevertheless, on-policy exploration against verifiable rewards remains the dominant recipe in this domain, a structural asymmetry we revisit in Section 11.

5.2. Process versus Outcome Supervision

Outcome rewards are cheap but coarse: a correct final answer can conceal flawed reasoning, and a single terminal scalar provides weak credit assignment over hundreds of steps. Uesato et al. (2022) first compared outcome- and process-based feedback on GSM8K, finding that outcome supervision matches final-answer accuracy but produces far more false-positive reasoning traces. Lightman et al. (2024) settled the question at scale: on MATH, a PRM trained on the 800K step-level labels of PRM800K substantially outperforms an ORM under best-of- N selection, and active learning multiplies the value of each human label. The bottleneck then became annotation cost, addressed by MathShepherd (P. Wang et al., 2024), which labels a step by the empirical probability that completions passing through it reach the correct answer, Monte Carlo process supervision without human annotators, and used the resulting PRM both for verification and for step-wise PPO. Setlur et al. (2024) reframed the target theoretically: effective process rewards should measure *progress*, the change in success probability under a prover policy, yielding process advantage verifiers

Table 2

Representative alignment methods. RM = explicit reward model required; Ref = reference model required; On = on-policy sampling during preference optimisation.

Method	Key idea	RM	Ref	On	Signature contribution
InstructGPT (Ouyang et al., 2022)	SFT → BT RM → PPO	✓	✓	✓	Established the RLHF standard
HH-RLHF (Bai, Jones, et al., 2022)	Iterated online RLHF	✓	✓	✓	Helpfulness–harmlessness trade-off
Sparrow (Glaese et al., 2022)	Rule RMs + evidence	✓	✓	✓	Decomposed, rule-conditioned rewards
Llama 2-Chat (Touvron et al., 2023)	Dual RMs + rejection sampling + PPO	✓	✓	✓	Open replication at 7–70B
Constitutional AI (Bai, Kadavath, et al., 2022)	Self-critique + RLAIF	✓	✓	✓	Harmlessness without human harm labels
RLAIF (H. Lee et al., 2024)	LLM preference labels	✓	✓	✓	Human-level alignment at lower cost
SALMON (Sun et al., 2024)	Instructable reward model	✓	✓	✓	Principle-steerable rewards
DPO (Rafailov et al., 2023)	Closed-form reparameterisation	—	✓	—	Eliminated RM and value network
IPO (Azar et al., 2024)	Bounded squared preference loss	—	✓	—	Fixed DPO overfitting
KTO (Ethayarajh et al., 2024)	Prospect-theoretic unpaired loss	—	✓	—	Learns from binary feedback
SimPO (Meng et al., 2024)	Length-normalised implicit reward	—	—	—	Reference-free, length-robust
ORPO (J. Hong et al., 2024)	Odds-ratio penalty inside SFT	—	—	—	Single-stage alignment
RainbowPO (Zhao et al., 2025)	Component-wise unification	—	✓	—	Systematic ablation of the design space
Online iterative DPO (Dong et al., 2024; Ye et al., 2024)	Sample-label-update loop	~	✓	✓	Mitigates offline distribution shift
XPO (Xie et al., 2024)	Optimistic exploration bonus	—	✓	✓	Provable sample efficiency
REBEL (Z. Gao et al., 2024)	Regression to relative rewards	~	✓	✓	PPO performance without clipping/critic

(PAVs) that improve both compute efficiency and exploration. ThinkPRM (Khalifa et al., 2025) then collapsed the data requirement by two orders of magnitude, fine-tuning a long-CoT model to verify each step generatively with only ~1K curated chains, and the GenRM line (Mahan et al., 2024) similarly moved verification into token space. The engineering loop of PRM construction, data generation, training, and deployment as verifier, reranker, or dense RL reward, is surveyed in detail by Zheng et al. (2025). In the corpus-wide statistics of Section 9.3, step-level credit assignment remains a minority practice (10.3%), concentrated almost entirely in this domain.

5.3. Self-Training and Bootstrapped Reasoning

A complementary off-policy lineage improves reasoning by iterated self-generation and filtered imitation. STaR (Zelikman et al., 2022) bootstraps rationales: the model generates chains of thought, keeps those that reach correct answers (rationalising failures from hints), and fine-tunes on the survivors, repeating until convergence. ReST^{EM} (A. Singh et al., 2024) formalised this loop as expectation–maximisation, generate (E-step), filter by binary correctness, fine-tune from the base model (M-step), and showed at PaLM scale that two to three iterations of self-generated data beat training on human-written solutions, with the largest gains at the largest model sizes. These methods can be read as policy iteration with a verifier-filtered replay buffer; they trade the exploration of on-policy RL for stability and simplicity, and

they seeded the data flywheels now embedded inside RLVR systems (DeepSeek-AI, 2025; Yang et al., 2024).

5.4. Verification and Test-Time Scaling

Reward models double as inference-time tools. Cobbe et al. (2021) introduced the paradigm: train a solution-level verifier and select the best of N sampled solutions, showing that verification scales more favourably with data than fine-tuning alone. Snell, Lee, Xu, and Kumar (2024) generalised the question to *compute-optimal* test-time scaling: given a fixed inference budget, adaptively choosing between best-of- N , beam search against a PRM, and sequential self-revision yields up to 4× efficiency gains over naive best-of- N , and on easy-to-medium problems test-time compute can substitute for a 14× larger model. These results connect the PRM literature of Section 5.2 to deployment economics and motivate the search-augmented agents of Section 6.

5.5. Formal Theorem Proving

Formal mathematics offers the purest reward signal in the corpus: the Lean 4 compiler either certifies a proof or it does not. DeepSeek-Prover-V1.5 (Xin et al., 2024) combines proof-assistant feedback with RLPAF (reinforcement learning from proof-assistant feedback) and RMaxTS, an intrinsic-reward Monte Carlo tree search over proof states, achieving state-of-the-art miniF2F results. Tool-integrated reasoning systems such as ToRA (Gou et al., 2024), which

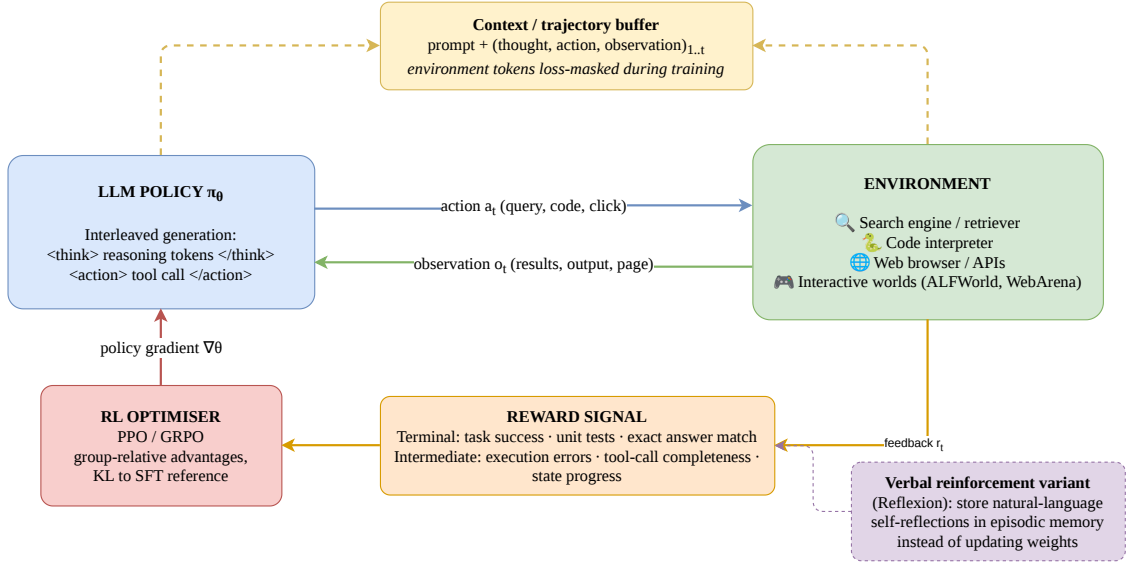


Figure 5: The agentic RL loop for tool-using LLMs. At each turn the policy interleaves reasoning tokens with tool actions (search queries, code execution, API calls); the environment returns observations that are appended to the context; and terminal (task success, unit tests) or intermediate (execution errors, state changes) environment feedback provides the reward used by the RL optimiser (typically PPO or GRPO). Verbal reinforcement variants (Shinn et al., 2023) store self-reflections in episodic memory instead of updating weights.

interleaves natural-language reasoning with symbolic library calls and repairs behaviour with output-space shaping, bridge this subsection to the tool-use domain that follows.

6. RL for Tool Use and Agentic Behaviour

The tool-use domain (19 papers, 17.8%) trains LLMs to act: to call APIs, execute code, search the web, and operate in interactive environments. Its distinguishing feature is that the reward signal comes from the *environment itself*, execution results, task completion, game state, rather than from human preferences or learned reward models. The canonical interaction loop is shown in Fig. 5. We organise the domain into agentic scaffolds without weight updates (Section 6.1), supervised tool learning (Section 6.2), end-to-end RL-trained agents (Section 6.3), and environments and benchmarks (Section 6.4).

6.1. Agentic Scaffolds without Weight Updates

Several foundational frameworks established the action space and feedback loops that later RL methods optimise, without themselves performing gradient updates. ReAct (Yao et al., 2023) interleaves free-form reasoning traces with environment actions, showing that the synergy between the two outperforms either alone on knowledge tasks (HotpotQA, FEVER) and interactive decision-making (ALFWorld, WebShop). Reflexion (Shinn et al., 2023) introduced *verbal reinforcement*: instead of updating weights, the agent converts scalar or binary environment feedback into natural-language self-reflections stored in episodic memory, improving subsequent trials, an in-context analogue of policy iteration. Voyager (G. Wang et al., 2023) demonstrated

lifelong learning in Minecraft through an automatic curriculum, an ever-growing skill library of verified executable code, and iterative prompting against execution feedback. PAL (L. Gao, Madaan, et al., 2023) offloaded computation to a Python interpreter, decomposing problems into programs whose execution provides exact answers, and CRITIC (Gou et al., 2024) generalised tool-grounded self-correction, showing that verification against external tools (search, interpreters, toxicity APIs) succeeds where intrinsic self-correction fails. These systems define the interfaces, thought–action–observation loops, executable action spaces, environment-derived feedback, that the RL-trained agents of Section 6.3 learn end-to-end.

6.2. Learning to Use Tools: Supervised and Self-Supervised Approaches

Toolformer (Schick et al., 2023) taught models *when* and *how* to call APIs through a self-supervised objective: candidate API calls are inserted into pre-training text and retained only if their results reduce future-token perplexity, requiring only a handful of demonstrations per tool. ToolLLM (Qin et al., 2023) scaled the paradigm to 16,464 real-world REST APIs, contributing the ToolBench instruction corpus, a depth-first-search decision-tree inference strategy that outperforms ReAct-style rollouts, and the ToolEval automatic evaluator. CodeAct (X. Wang et al., 2024) consolidated the action space itself, arguing that executable Python is a strictly more expressive action format than JSON tool calls, composition, control flow, and variable reuse come for free, and reporting up to 20% higher success rates. These supervised foundations supply both the data and the action formats on which agentic RL now builds.

6.3. End-to-End RL-Trained Agents

The 2025 wave applies RLVR-style training directly to multi-turn tool interaction. Search-R1 (Jin et al., 2025) trains a policy that autonomously issues search queries during step-by-step reasoning, using retrieved-token masking for stable optimisation and a simple outcome reward, yielding 20–40% relative gains over retrieval-augmented baselines. ReTool (J. Feng et al., 2025) applies outcome-driven RL to code-interpreter use for mathematical reasoning, discovering emergent behaviours such as code self-correction and achieving 67% on AIME 2024 with far fewer steps than text-only RL; ToRL (X. Li et al., 2025) reports the same emergent tool-use competencies when scaling tool-integrated RL from base models. On the framework side, Agent-R1 (Cheng et al., 2025) extends the MDP formulation to end-to-end multi-turn agent training with process rewards for tool calls, AgentRL (H. Zhang et al., 2025) contributes the systems infrastructure, asynchronous generation, containerised environment clusters, cross-policy sampling, for scaling multi-turn multi-task agentic RL, and RLFactory (Chai, Song, Wei, et al., 2025) provides a plug-and-play post-training framework that decouples environment construction from training logic. AGILE (P. Feng et al., 2024) frames the complete agent, memory, tools, expert-consultation, reflection, as a single RL policy, showing that a trained 13B agent can surpass GPT-4 on complex QA, while ARTIST (J. Singh, Magazine, Pandya, & Nambi, 2025) unifies agentic reasoning and tool integration under outcome-based GRPO. RLTR (Z. Li, Hu, & Wang, 2025) isolates a key credit-assignment insight: decoupling planning from answer generation and rewarding *tool-call completeness* alone improves end-to-end response quality, sidestepping the noisy joint signal of full-response evaluation.

Across these systems three design regularities recur: (1) environment feedback is used directly as reward, reducing dependence on learned reward models and their hacking pathologies, though sparse task-success signals, exploitable simulators, and under-specified unit tests remain proxy rewards in their own right and are still vulnerable to specification gaming; (2) loss masking of environment-generated tokens is essential for stability; and (3) GRPO-style group-relative advantages have displaced PPO's value network, mirroring the reasoning domain. Conspicuously absent, as Section 9.5 documents, is safety evaluation: none of the corpus's tool-use papers assesses harmful tool-call behaviour, despite these agents having the most direct real-world affordances.

6.4. Environments and Benchmarks

Progress in this domain is inseparable from its environments. Mind2Web (Deng et al., 2023) contributed the first dataset for *generalist* web agents, 2,350 tasks over 137 real websites, exposing the brittleness of models trained on synthetic sites. WebArena (Zhou et al., 2024) provides a self-hostable, functionally realistic multi-site environment with execution-based evaluation, on which early GPT-4 agents achieved only 14.4% end-to-end success against 78% human

performance, quantifying the gap that RL-trained agents now target. Together with WebShop, ALFWorld, ToolBench (Qin et al., 2023), and ToolEval, these environments supply the verifiable, resettable reward sources that make agentic RL tractable; their fragmentation across papers, however, contributes to the benchmark-comparability problems analysed in Section 9.5.

7. RL for Safety and Robustness

The safety domain (18 papers, 16.8%) comprises 14 systems that train attacker or defender policies with RL, used both *offensively*, training attacker policies that probe target models, and *defensively*, constraining or hardening the trained policy, together with four contextual studies (crowd-sourced red-team data collection, gradient-based attacks, and safety-focused supervised fine-tuning) that frame and calibrate those RL methods; Table 7 separates the two groups. It is the only domain in which every paper includes dedicated safety evaluation. We cover automated red-teaming (Section 7.1), constrained and rule-guided safety training (Section 7.2), and adversarial attacks and robustness (Section 7.3).

7.1. RL-Based Automated Red-Teaming

Perez et al. (2022) pioneered the offensive use: a red-team LM is trained by zero/few-shot generation, supervised learning, and finally RL against a harmfulness classifier reward to elicit offensive outputs from a target chatbot, uncovering tens of thousands of successful attacks. The central failure mode of this recipe is *mode collapse*: RL maximizes attack success at the cost of diversity. Subsequent work is largely a sequence of diversity mechanisms. Curiosity-driven red-teaming (Z.-W. Hong et al., 2024) adds novelty and entropy bonuses, recovering diversity while preserving effectiveness, even eliciting toxic responses from RLHF-hardened models. Rainbow Teaming (Samvelyan et al., 2024) recasts attack generation as quality diversity search over a MAP-Elites archive spanning risk categories and attack styles, producing hundreds of diverse successful prompts and transferring across models and modalities. GFlowNet fine-tuning (S. Lee et al., 2025) samples attacks proportionally to reward rather than maximising it, and its smoothed re-training stage yields attacks that transfer to unseen target models. Explore Establish Exploit (Casper, Lin, Kwon, Culp, & Hadfield-Menell, 2023) attacks *from scratch*, constructing the harmfulness reward itself from human labels on the target's outputs so that red-teaming does not presuppose a pre-existing classifier. Active Attacks (Yun et al., 2025) closes the loop with adaptation: the victim is periodically fine-tuned against current attacks, forcing the attacker to explore new regions and yielding order-of-magnitude improvements in cross-attack success. RLbreaker (X. Chen, Nie, Guo, & Zhang, 2024) shows the same machinery driving jailbreak search directly, modelling jailbreaking as a deep RL-guided search that outperforms genetic-algorithm baselines. On the human side, Ganguli et al. (2022) released 38,961 crowd-sourced red-team attacks

with a taxonomy of harms, documenting that plain LMs and RLHF models differ sharply in attackability and providing the data substrate for many classifier rewards above.

7.2. Constrained and Rule-Guided Safety Training

Defensively, Safe RLHF (Dai et al., 2023) formalises the helpfulness–harmlessness tension as a constrained MDP: separate reward (helpfulness) and cost (harmlessness) models are trained from decoupled annotations, and a Lagrangian dual variable dynamically balances the two objectives during PPO, improving both axes simultaneously where naive multi-objective mixing trades them off. Rule-Based Rewards (Mu et al., 2024) decompose safety policy into fine-grained propositions (e.g., “refusals should contain a brief apology”) scored by an LLM grader, combining these with the helpfulness RM to hit precise behavioural targets with minimal human data; the same mechanism drives diversity-aware multi-step red-teaming in Beutel, Xiao, Heidecke, and Weng (2024). MART (Ge et al., 2024) couples attack and defence in multi-round adversarial self-play: the attacker generates, the victim is safety-fine-tuned on flagged failures, and repeat, reducing violation rates up to 84.7% while preserving helpfulness. The instruction hierarchy (Wallace et al., 2024) teaches models to privilege system over user over tool-output instructions, addressing prompt-injection at the training level, and Safety-Tuned LLaMAs (Bianchi et al., 2024) documents the small-data regime: a few hundred safety examples substantially improve safety, while excessive safety data induces exaggerated refusal, the over-refusal side of the alignment tax.

7.3. Adversarial Attacks and Robust Training

Optimisation-based attacks calibrate how robust these defences really are. GCG (Zou et al., 2023) showed that greedy coordinate-gradient search over adversarial suffixes produces *universal and transferable* jailbreaks optimized on open-weight models yet effective against proprietary systems, establishing that alignment achieved by RLHF alone is brittle to white-box optimization pressure. Targeted latent adversarial training (LAT) (Sheshadri et al., 2025) responds by perturbing hidden activations rather than input tokens, hardening models against classes of failures without requiring instances of the triggering inputs; it improves jailbreak robustness, backdoor removal, and unlearning at minimal capability cost. Together with the red-team-then-harden loops above, these works define an escalating attacker–defender co-evolution whose equilibrium properties remain an open question (Section 11).

8. RL for Continual Improvement

The smallest and most nascent domain addresses a deployment reality: aligned models must keep improving after release, from their own outputs and from non-stationary feedback streams, without forgetting. Three mechanisms have emerged. **Self-play**: SPIN (Z. Chen et al., 2024) trains the model against its previous iteration; the current policy learns to distinguish (and prefer) human data over its

own prior generations, provably converging when the policy matches the data distribution and improving without any new human annotation. **Iterative online preference optimization**: the online RLHF workflows of Dong et al. (2024) and Ye et al. (2024) (Section 4.4) constitute the practical machinery for continual updating, regenerating preference data from the evolving policy at each iteration; OFS-DPO (Qi et al., 2024) adds explicit forgetting control through fast–slow module competition. **Parameter-space regularization**: the Online Merging Optimizer (Lu et al., 2024) interpolates gradients with the task vector between the RLHF policy and the SFT initialization at every step, boosting alignment reward while suppressing alignment tax across model families and RLHF algorithms. The open problems of this domain reward staleness, capability retention, and lifelong safety are among the least studied in the corpus and reappear in Section 11.

9. Cross-Cutting Analysis and Trends

Whereas Sections 4–8 examined each domain in depth, this section reads the corpus *across* domains. All statistics refer to the 107 curated studies described in Section 1.3; because the corpus is purposive, they describe this curated sample and are indicative of the methodological landscape rather than population estimates. Every table and percentage in this section is computed programmatically from the released study-level evidence table by a single analysis script with automated assertions that all marginal totals reconcile. Section 9.1 characterises the corpus; Section 9.2 cross-tabulates the taxonomy; Sections 9.3–9.7 analyse algorithms, reward design, evaluation, failure modes, and reporting practices in turn.

9.1. Corpus Characteristics

9.1.1. Temporal Distribution

Table 3 summarises the year-by-year distribution of the 107 studies, and Fig. 6 breaks the trend down by functional domain. The corpus is heavily weighted toward recent years, with 2024 alone contributing 48 of the selected papers (44.9%) and the 2024–2026 period accounting for 70.1%. Because selection was purposive and deliberately domain-balanced (Section 1.3), these figures describe the curated sample and reflect the recency of the underlying literature only qualitatively; they are not estimates of publication rates or growth curves. The recency weighting tracks three concurrent developments: the proliferation of DPO variants (J. Hong et al., 2024; Meng et al., 2024; Rafailov et al., 2023), reasoning-via-RL methods (ByteDance Seed and Tsinghua AIR, 2025; DeepSeek-AI, 2025; Shao et al., 2024), and safety-specific RL algorithms (Dai et al., 2023; Perez et al., 2022; Samvelyan et al., 2024). The pre-2022 period is sparsely populated, reflecting the fact that large-scale RLHF research only became practical with the release of InstructGPT (Ouyang et al., 2022) and the subsequent open-weight model wave.

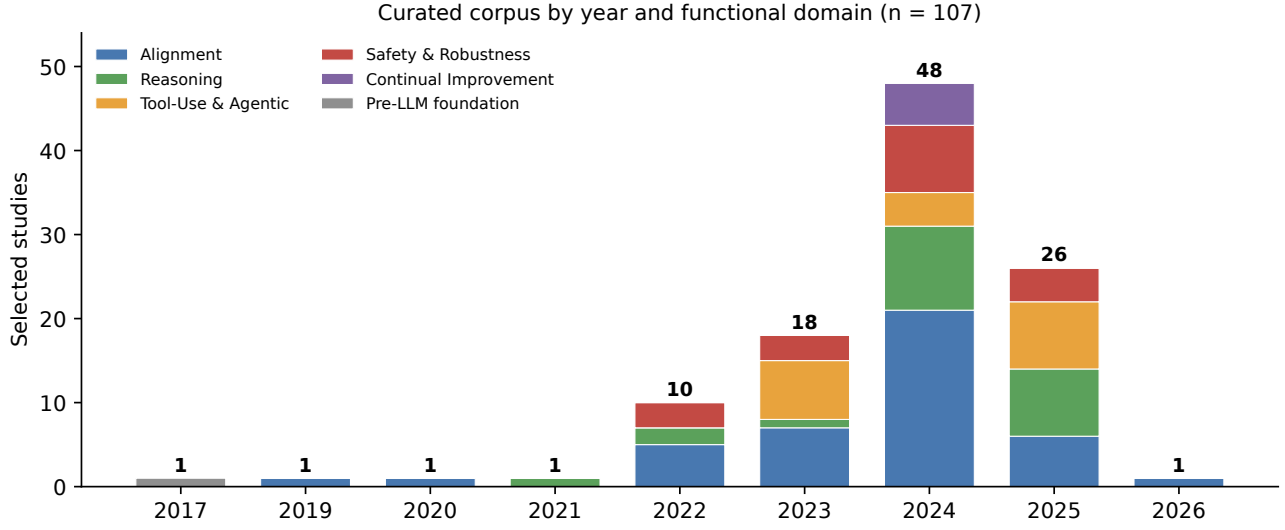


Figure 6: Curated corpus by year and functional domain (2017–2026, $n = 107$), generated programmatically from the released evidence table. The 2024–2026 period accounts for 70.1% of the selected studies. Because the corpus is purposive and domain-balanced, bar heights describe the curated sample, not field-wide publication rates. Alignment dominates across all years, while Reasoning, Tool-Use, and Safety enter primarily from 2022–2023 onward; the single 2026 entry is the latest version of the evolving survey of Z. Wang et al. (2026).

Table 3

Temporal distribution of the curated studies ($n = 107$). Percentages are rounded to one decimal place and may not sum to exactly 100.

Year	Count	%
2017	1	0.9
2019	1	0.9
2020	1	0.9
2021	1	0.9
2022	10	9.3
2023	18	16.8
2024	48	44.9
2025	26	24.3
2026	1	0.9
Total	107	100.0

9.1.2. Venue and Publication Type

The distribution of studies across venue tiers is presented in Table 4. After the final reference audit, in which eight studies previously coded at their arXiv versions were re-coded at their subsequently verified conference versions (logged in the codebook), peer-reviewed papers account for 53 of 107 (49.5%), with NeurIPS the most represented venue (16 papers), followed by ICLR (12). Preprints still constitute a slim majority of the corpus (50.5%), a share that reflects the rapid dissemination culture characteristic of LLM research, in which high-impact works such as DeepSeek-R1 (DeepSeek-AI, 2025) and DAPO (ByteDance Seed and Tsinghua AIR, 2025) are first released on arXiv before formal peer review; roughly half of the influential studies surveyed here had not undergone formal peer review at the time of coding.

9.1.3. Affiliation Type

Table 5 reports the author affiliation breakdown. Hybrid affiliations, industry-academic collaborations, represent the plurality at 42.1%, reflecting the deeply collaborative nature of RL-for-LLM research. Key industry contributors include OpenAI (OpenAI, 2023, 2024; Ouyang et al., 2022), DeepSeek (DeepSeek-AI, 2025; Shao et al., 2024; Xin et al., 2024), Anthropic (Bai, Jones, et al., 2022; Bai, Kadavath, et al., 2022), Meta (Schick et al., 2023; Touvron et al., 2023), and Alibaba (Yang et al., 2024; Z. Zhang et al., 2025). Purely academic contributions (19.6%) are the smallest category, underscoring that the computational demands of RL-for-LLM research increasingly necessitate industrial resources or partnerships.

9.1.4. Model Scale

The parameter-scale distribution is shown in Table 6. The 1B–10B bracket dominates at 58.9%, reflecting the

Table 4Venue tier distribution ($n = 107$).

Tier	Venues	Count	%
Tier 1	NeurIPS (16), ICLR (12), ICML (9), ACL (2), EMNLP (5), NAACL (1), ACM Comp. Surv. (1)	46	43.0
Tier 2	TMLR (3), AISTATS (1), ACL Findings (2), COLM (1)	7	6.5
Preprint	arXiv (54)	54	50.5
Total		107	100.0

Table 5Author affiliation distribution ($n = 107$).

Affiliation	Count	%
Industry	41	38.3
Academic	21	19.6
Hybrid	45	42.1
Total	107	100.0

accessibility of open-weight models such as LLaMA 2 (Touvron et al., 2023), Mistral, and Qwen (Yang et al., 2024). Within this bracket, the 7B scale is by far the most common choice, establishing it as the *de facto* standard experimental size for RL-for-LLM research. The largest model in the corpus is DeepSeek-R1 at 671B parameters in a Mixture-of-Experts (MoE) configuration (DeepSeek-AI, 2025). Notably, 27.1% of papers do not report model scale, contributing to the reporting concerns discussed in Section 9.7. Because papers experimenting at multiple scales are counted in each applicable bracket, the bracket counts sum to 141 across the 107 studies; Table 6 should therefore be read as a multiple-response table rather than a distribution. The functional-domain distribution of all 107 studies is illustrated in Fig. 7.

9.2. Taxonomy Cross-Tabulation

The cross-tabulation of the two taxonomy axes, functional purpose and optimisation architecture, across all 107 studies is shown in Table 7; the table is regenerated directly from the study-level evidence table, and all row and column totals reconcile to 107. On-policy methods (50 papers, 46.7%) dominate across all domains, particularly in Safety (14/18 = 77.8%), where online interaction with the environment is essential for adversarial training (Z.-W. Hong et al., 2024; Perez et al., 2022). Reward-free methods (17 papers) are concentrated exclusively in the Alignment domain, with *zero* applications to Reasoning within this corpus, a notable asymmetry. In the broader literature, early adaptations of offline preference objectives to reasoning have appeared (e.g., Step-DPO (Lai et al., 2024) and Full-Step-DPO (H. Xu et al., 2025)), but they remain the exception rather than the rule. The N/A column (25 papers) includes survey papers (Chaudhari et al., 2025; Xiao et al., 2025; K. Zhang et al., 2025; Zheng et al., 2025), SFT-based tool-use methods (Qin et al., 2023; Schick et al., 2023), prompting frameworks (Shinn et al., 2023; G. Wang et al., 2023; Yao et al., 2023), and self-play fine-tuning (Z. Chen et al., 2024). (The N/A row of Table 8 contains exactly the same 25 non-training studies as this column; the nine trained systems whose procedures fall outside the established named families are

itemised in Table 8 as explicit agentic, iterative-RLHF, and pre-LLM subfamilies.)

The three-way relationship between functional domains, optimisation architectures, and algorithm families is visualised in Fig. 8, which reveals the dominant flow paths: Alignment → On-Policy → PPO, Alignment → Reward-Free → DPO family, and Reasoning → On-Policy → GRPO/DAPO.

9.3. Algorithms and Optimisation Practices

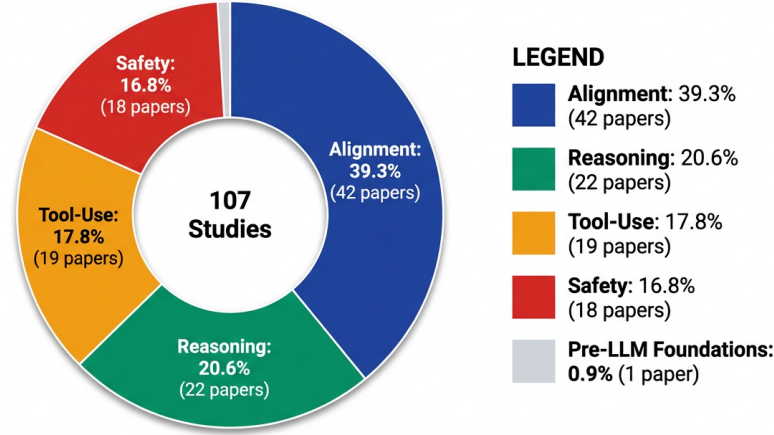
9.3.1. Algorithm Distribution

The full algorithm family distribution is presented in Table 8. PPO (Ouyang et al., 2022) remains the most prevalent single algorithm family at 18.7% (20 of 107 papers), but its dominance has eroded over time. To compare like with like, we measure the trend in a single unit, *PPO-style actor-critic optimiser usage among RL-trained systems*, recorded as a dedicated field in the released evidence table: among the 11 trained systems in the 2020–2022 subset (the subset holds 12 studies in total, one of which trains no policy), 9 (82%) use PPO-style optimisation (this usage-based count includes two systems whose primary family lies elsewhere; both are documented in the codebook), whereas corpus-wide only 35 of the 82 trained systems (42.7%) do. Because the early period contributes few studies, this comparison should still be read as indicative of a trend rather than as a precise year-on-year prevalence estimate. This decline is driven by the emergence of reward-free methods (DPO family: 14.0%) (J. Hong et al., 2024; Meng et al., 2024; Rafailov et al., 2023), lightweight on-policy alternatives (GRPO/DAPO: 6.5%) (ByteDance Seed and Tsinghua AIR, 2025; Shao et al., 2024) that eliminate the value network, and domain-specific algorithms for safety (Z.-W. Hong et al., 2024; Perez et al., 2022; Samvelyan et al., 2024) and formal theorem proving (Xin et al., 2024). All 82 trained systems are assigned to a family in Table 8, 73 to established named families and 9 to the explicit agentic, iterative-RLHF, and pre-LLM subfamilies introduced there; among these 82, on-policy families account for $50/82 = 61.0\%$. The long tail of the distribution, 14 algorithm families each appearing only once, illustrates the substantial methodological diversification within the curated corpus.

Table 6

Model parameter scale ($n = 107$). This is a multiple-response table: papers experimenting at several scales are counted once in each applicable bracket, so counts sum to more than 107 and percentages (share of the 107 studies) do not form a 100% distribution. Brackets are mutually exclusive; MoE models are coded by total (not active) parameter count.

Scale Bracket	Count	% of studies
< 1B	5	4.7
1B – 10B	63	58.9
>10B – 70B	32	29.9
>70B – 600B	11	10.3
> 600B (MoE)	1	0.9
N/A / NR	29	27.1



Period: 2017–2025. Alignment is the dominant domain at 39.3%.

Figure 7: Functional domain distribution of the curated studies ($n = 107$). Alignment remains the largest domain at 39.3% (42 papers), followed by Reasoning at 20.6% (22 papers), Tool-Use at 17.8% (19 papers), Safety at 16.8% (18 papers), and Continual Improvement at 4.7% (5 papers). Pre-LLM foundations account for 0.9% (1 paper), so all slices sum to 107 studies. The comparable representation of the four largest LLM domains is a design property of the curated evidence map, coverage criterion (4) of Section 1.3 deliberately balanced domains, and should not be read as an empirical property of the literature.

9.3.2. Credit Assignment Granularity

Table 9 reports how RL credit is distributed across the generation process. Sequence-level credit assignment dominates at 61.7%, reflecting the prevalence of outcome-based rewards where the entire model response receives a single scalar signal. Step-level process supervision accounts for 10.3% of studies and is concentrated almost exclusively in the Reasoning domain, driven by the PRM800K dataset (Lightman et al., 2024), Math-Shepherd (P. Wang et al., 2024), process-advantage verifiers (PAVs) (Setlur et al., 2024), and ThinkPRM (Khalifa et al., 2025). The sole proof-step-level entry represents DeepSeek-Prover-V1.5 (Xin et

al., 2024), where the Lean 4 compiler provides ground-truth step verification, the only instance in the corpus where each intermediate reasoning step receives a formally verified reward signal. The dominance of sequence-level credit suggests that the field has not yet systematically addressed the challenge of fine-grained feedback, despite growing evidence that process supervision yields superior reasoning accuracy (Lightman et al., 2024; Uesato et al., 2022).

9.3.3. Compute Budget Reporting

As shown in Table 10, only 6 of 107 papers (5.6%) report compute budgets in the form of GPUxhours or FLOPs.

Table 7

Taxonomy: functional domain $\times$ optimisation architecture ($n = 107$).

Domain	On-Pol.	Off-Pol.	Hybrid	Rew-Free	N/A	Total
Alignment	14	1	4	17	6	42
Reasoning	11	7	0	0	4	22
Tool-Use	9	0	0	0	10	19
Safety	14	0	0	0	4	18
Continual	1	0	3	0	1	5
Pre-LLM	1	0	0	0	0	1
Total	50	8	7	17	25	107

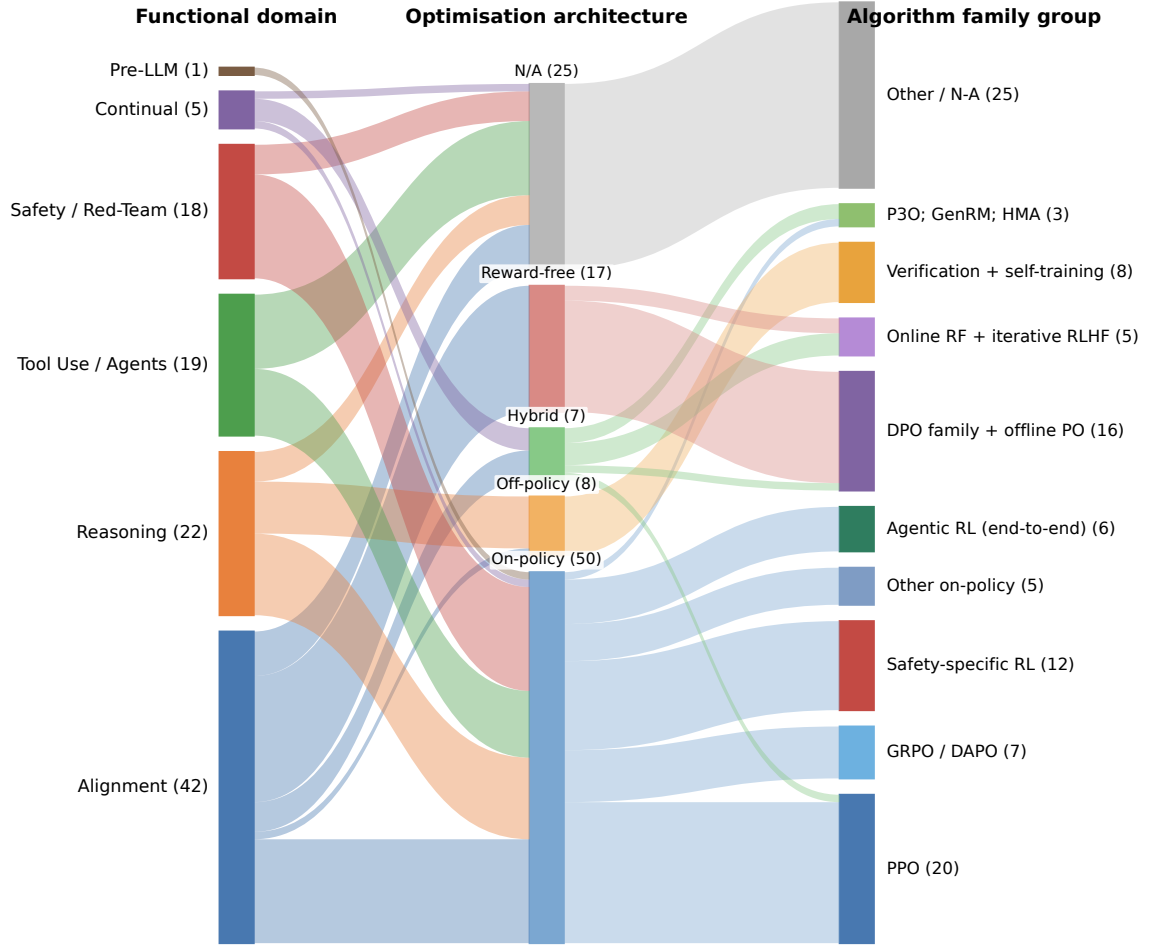


Figure 8: Alluvial (Sankey-style) diagram of the three-way flow between functional domains (left), optimisation architectures (centre), and algorithm family groups (right) across the 107 studies; the figure is generated programmatically from the released evidence table, and ribbon width is proportional to the number of studies. Key observations: PPO is fed primarily by Alignment and Safety through on-policy architectures; the DPO family is fed exclusively by Alignment through reward-free methods; GRPO/DAPO is fed by Reasoning through on-policy methods; the end-to-end agentic-RL subfamilies (PPO- and GRPO-based) flow from the Tool-Use domain; and the Other/N-A group now contains only the 25 non-training studies (SFT-based, prompting-based, benchmark, dataset, and survey papers).

Notable exceptions include Tulu 3 (Lambert et al., 2025) and DAPO (ByteDance Seed and Tsinghua AIR, 2025), both of which provide detailed resource accounting. This constitutes one of the most severe reporting gaps identified in this survey. Computing cost is essential for replication and has direct implications for the accessibility of RL-for-LLM research: without knowing whether a method requires 100 or 10,000 GPU-hours, practitioners cannot assess feasibility, and the community cannot make meaningful cost-benefit comparisons across algorithms.

9.4. Reward Design and Feedback Mechanisms

9.4.1. Reward Signal Types

The full reward signal taxonomy is presented in Table 11 and visualised in Fig. 9, which shows the 18 types as horizontal bars grouped into five families; the complete

per-study type mapping is released with the evidence table. The corpus encompasses **18 distinct reward types**, the largest reward-type taxonomy compiled in the RL-for-LLM literature to date. Reward design has become domain-specific: learned Bradley-Terry models and implicit DPO signals dominate in alignment (Bai, Jones, et al., 2022; Ouyang et al., 2022; Rafailov et al., 2023); process reward models (PRMs) and rule-based verification are the primary choices for reasoning (DeepSeek-AI, 2025; Lightman et al., 2024; P. Wang et al., 2024); environment feedback from API execution, code compilation, and game environments serves tool use (Schick et al., 2023; G. Wang et al., 2023; Yao et al., 2023); and classifier-based rewards drive safety red-teaming (Z.-W. Hong et al., 2024; Perez et al., 2022). This domain-specificity implies that reward engineering has matured from a generic pipeline component into a first-class design choice requiring domain expertise.

Table 8

Algorithm family distribution ($n = 107$). Each study is assigned to exactly one *primary* algorithm family (its principal training contribution); studies employing multiple algorithms are coded by the algorithm central to their contribution, so rows are mutually exclusive and counts sum to 107. All 82 trained systems are assigned to a family: nine systems trained with bespoke or framework-specific procedures are itemised as explicit subfamilies (end-to-end agentic RL, PPO- and GRPO-based; the online iterative RLHF workflow; and pre-LLM preference RL), so the N/A row (25, 23.4%) contains exactly the non-training studies of the N/A architecture column in Table 7.

Algorithm Family	Count	%
<i>On-policy reward-model-based:</i>		
PPO (alignment + reasoning + agent + safety) (Dai et al., 2023; Ouyang et al., 2022; Touvron et al., 2023)	20	18.7
GRPO / DAPO (ByteDance Seed and Tsinghua AIR, 2025; DeepSeek-AI, 2025; Shao et al., 2024)	7	6.5
A2C (Glaese et al., 2022; Menick et al., 2022)	2	1.9
RLPAF (Xin et al., 2024)	1	0.9
RL red-team (classifier reward) (Perez et al., 2022)	2	1.9
Curiosity-driven RL (CRT) (Z.-W. Hong et al., 2024)	1	0.9
GFlowNet fine-tuning (S. Lee et al., 2025)	1	0.9
QD open-ended search (Rainbow) (Samvelyan et al., 2024)	1	0.9
Iterative adversarial DPO-style (APL) (Y. Wang, Zhou, Zhai, et al., 2025)	1	0.9
Multi-step RL with RBR (Diverse Red-Team) (Beutel et al., 2024)	1	0.9
RL safety fine-tuning (MART) (Ge et al., 2024)	1	0.9
RL with LLM-graded RBR (Mu et al., 2024)	1	0.9
Targeted latent adversarial training (LAT) (Sheshadri et al., 2025)	1	0.9
RL red-team from scratch (E3) (Casper et al., 2023)	1	0.9
RL adaptive red-teaming (Active Attacks) (Yun et al., 2025)	1	0.9
Online Merging Optimizer (RLHF + model merging) (Lu et al., 2024)	1	0.9
End-to-end agentic RL (PPO-based) (Cheng et al., 2025; J. Feng et al., 2025; P. Feng et al., 2024; Z. Li et al., 2025)	4	3.7
End-to-end agentic RL (GRPO-based) (Chai et al., 2025; X. Li et al., 2025)	2	1.9
<i>Off-policy / verification:</i>		
BoN / verification / search (Cobbe et al., 2021; Khalifa et al., 2025; Snell et al., 2024)	4	3.7
Self-training (STaR/ReST ^{EM}) (A. Singh et al., 2024; Zelikman et al., 2022)	2	1.9
<i>Reward-free (offline DPO-family):</i>		
DPO + variants + analysis (Ethayarajh et al., 2024; J. Hong et al., 2024; Meng et al., 2024; Rafailov et al., 2023)	15	14.0
POWER (robust reward weighting) (Rashidinejad & Tian, 2024)	1	0.9
<i>Online reward-free:</i>		
XPO; REBEL; OFS-DPO (Z. Gao et al., 2024; Qi et al., 2024; Xie et al., 2024)	3	2.8
Online iterative RLHF workflow (Dong et al., 2024; Ye et al., 2024)	2	1.9
<i>Other / N/A:</i>		
PRM/ORM study (Lightman et al., 2024; Uesato et al., 2022)	2	1.9
P3O; GenRM; HMA (Lin et al., 2024; Mahan et al., 2024; Wu et al., 2023)	3	2.8
Pre-LLM preference RL (policy gradient) (Christiano et al., 2017)	1	0.9
N/A (non-training studies: SFT, prompting, gradient attack, benchmark, dataset, manual red-teaming, survey)	25	23.4
Total	107	100.0

Table 9

Credit assignment granularity ($n = 107$).

Granularity	Count	%
Sequence-level (outcome)	66	61.7
Step-level (process)	11	10.3
Proof-step level (formal)	1	0.9
Both studied	3	2.8
N/A / NR	26	24.3

9.4.2. Feedback Sources

Table 12 reports the feedback source distribution across the corpus. A key finding is that non-human sources, automatic/rule-based (19), environment feedback (11), AI-generated (8), self-generated (8), and synthetic (3), together 49 source usages spanning 45 distinct studies (42.1%), substantially exceed human annotation (expert annotators plus crowdworkers: 13 studies, 12.1%). Because a study can draw on several feedback sources, the table is multi-label and its counts sum to more than 107. This shift reflects the field's systematic movement away from costly human feedback, enabled by RLAIIF (Bai, Kadavath, et al., 2022; H. Lee et al., 2024), self-play (Z. Chen et al., 2024), and automatic verification (ByteDance Seed and Tsinghua AIR, 2025; DeepSeek-AI, 2025). Open-source and existing datasets (21.5%) form the largest single category, reflecting the growing ecosystem of publicly available preference data. The

Table 10Compute budget reporting ($n = 107$).

Compute Reported	Count	%
Yes	6	5.6
No	101	94.4

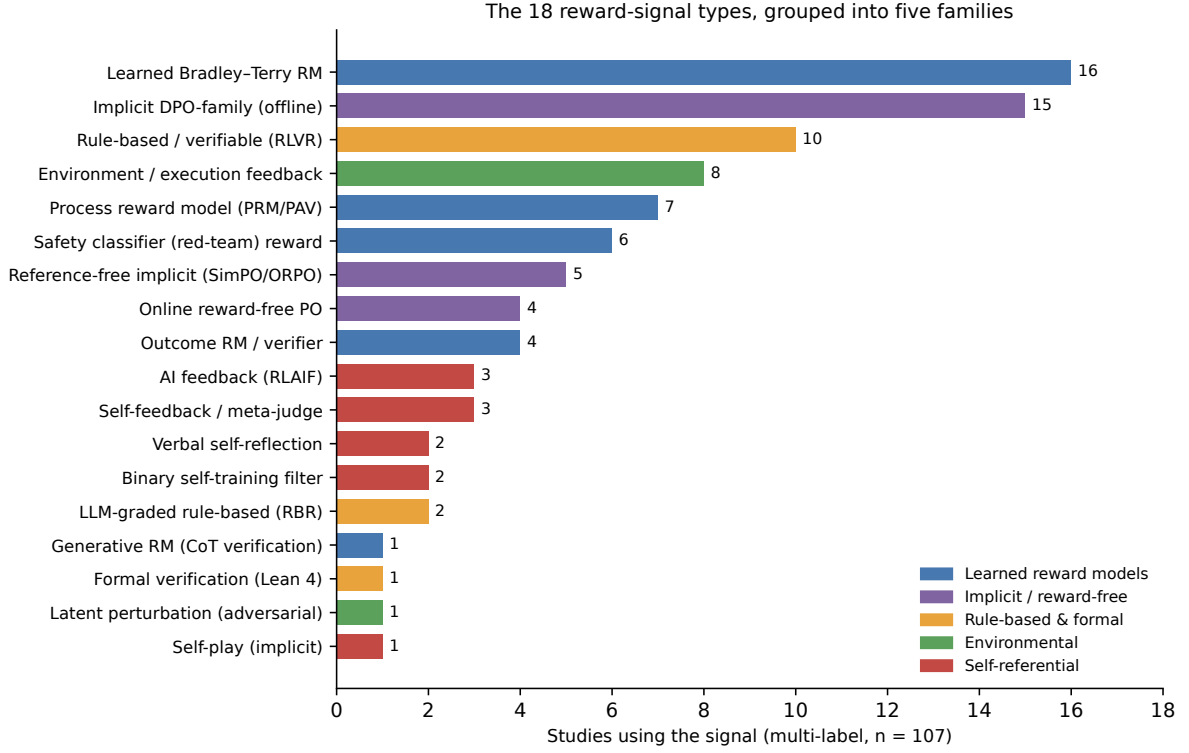


Figure 9: The 18 reward-signal types identified in the corpus, ordered by prevalence (multi-label study counts, $n = 107$) and coloured by family: learned reward models, implicit/reward-free signals, rule-based and formal signals, environmental signals, and self-referential signals. The figure is generated programmatically from the released evidence table; the full inventory with representative methods appears in Table 11.

implications are twofold: scalability has improved dramatically, but the quality and diversity of non-human feedback remain open questions that warrant systematic investigation.

9.4.3. Reward Failure Reporting

As shown in Table 13, just over half (52.3%) of studies explicitly report reward-related failure modes. Among those that do, notable contributions include the reward model overoptimisation scaling laws of L. Gao, Schulman, and Hilton (2023), which quantify the relationship between KL divergence and proxy reward degradation; the reward hacking analysis in Bai, Jones, et al. (2022), which documents how models learn to exploit reward model weaknesses; and the mode collapse findings in RL red-teaming (Z.-W. Hong et al., 2024; Yun et al., 2025), where attack models converge on a narrow set of successful jailbreak strategies. The 47.7% of studies that omit failure reporting likely underreport reward pathologies, given that reward hacking is the most prevalent failure mode in the corpus (28.0%; see Section 9.6).

9.5. Evaluation Practices and Benchmarks

9.5.1. Evaluation Paradigm

Table 14 shows the evaluation paradigm distribution. Automatic evaluation dominates at 72.0%, reflecting the community's reliance on benchmark-based assessment over human judgment. The hybrid category (24.3%) comprises studies that complement automatic metrics with genuine human preference or expert evaluation (Bai, Jones, et al., 2022; Dang et al., 2024; Ouyang et al., 2022); LLM-as-judge protocols such as AlpacaEval and MT-Bench grading are counted as automatic evaluation, not human evaluation, throughout. The four N/A entries are the comparator surveys, which evaluate no system. Strikingly, only InstructGPT (Ouyang et al., 2022) and Sparrow (Glaese et al., 2022) rely primarily on human evaluation, both among the earliest papers in the corpus. This shift suggests that the field has largely outsourced evaluation quality to benchmark design rather than direct human assessment, raising questions about whether current benchmarks adequately capture the nuances of model behaviour that human evaluators would detect.

Table 11

Reward signal taxonomy ($n = 107$); 18 distinct reward types identified. This is a multi-label table: studies that combine several reward signals are counted once under each applicable type, so counts sum to more than 107 and percentages denote the share of the 107 studies using each type. The study-to-type mapping is released with the evidence table.

Reward Type	Count	%
Implicit DPO-family (offline) (Ethayarajh et al., 2024; J. Hong et al., 2024; Meng et al., 2024; Rafailov et al., 2023)	15	14.0
Learned BT (sequence-level) (Bai, Jones, et al., 2022; Ouyang et al., 2022; Touvron et al., 2023)	16	15.0
Process Reward Model (PRM/GenPRM/PAV) (Khalifa et al., 2025; Lightman et al., 2024; Setlur et al., 2024; P. Wang et al., 2024)	7	6.5
Rule-based / verifiable (ByteDance Seed and Tsinghua AIR, 2025; DeepSeek-AI, 2025; Shao et al., 2024)	10	9.3
LLM-graded rule-based (RBR) (Beutel et al., 2024; Mu et al., 2024)	2	1.9
Latent perturbation (adversarial) (Sheshadri et al., 2025)	1	0.9
ORM / verifier (Cobbe et al., 2021; Uesato et al., 2022)	4	3.7
Generative RM (CoT verification) (Mahan et al., 2024)	1	0.9
Reference-free implicit (J. Hong et al., 2024; Meng et al., 2024)	5	4.7
Online reward-free (Z. Gao et al., 2024; Xie et al., 2024)	4	3.7
Environment feedback (tool-use) (Schick et al., 2023; G. Wang et al., 2023; Yao et al., 2023)	8	7.5
Verbal / self-reflection (Shinn et al., 2023)	2	1.9
AI-generated (RLAIF) (Bai, Kadavath, et al., 2022; H. Lee et al., 2024)	3	2.8
Self-feedback / meta-judge (Sun et al., 2024; Wu et al., 2024)	3	2.8
Self-play (implicit self-improvement) (Z. Chen et al., 2024)	1	0.9
Binary self-training (A. Singh et al., 2024; Zelikman et al., 2022)	2	1.9
Formal verification (Xin et al., 2024)	1	0.9
Classifier-based (red-team reward) (Z.-W. Hong et al., 2024; Perez et al., 2022)	6	5.6
NR / N/A	21	19.6

Table 12

Feedback source distribution ($n = 107$). Multi-label table: studies drawing on several feedback sources are counted once per source, so counts sum to more than 107; percentages denote the share of studies using each source.

Source	Count	%
Open-source / existing data	23	21.5
Automatic / rule-based	19	17.8
Environment feedback	11	10.3
Expert annotators	8	7.5
AI feedback (LLM-generated)	8	7.5
Self-generated / self-training	8	7.5
Crowdworkers	5	4.7
Synthetic (controlled)	3	2.8
NR / N/A	26	24.3

9.5.2. Benchmark Usage

The most frequently used benchmarks are listed in Table 15. MATH (Hendrycks et al., 2021) (19 papers) and GSM8K (Cobbe et al., 2021) (16 papers) have overtaken alignment benchmarks as the most-used evaluation datasets, reflecting the shift toward reasoning as a primary research focus. On the alignment side, AlpacaEval 2 (14 papers), MT-Bench (13), and Arena-Hard (9) form the standard evaluation triplet. The corpus spans over 50 distinct benchmarks with no standardised suite per domain, representing

significant benchmark fragmentation that impedes cross-study comparison. This fragmentation is particularly acute in safety evaluation, where custom red-team evaluations (5 papers), JailbreakBench (3), and HarmBench (3) are used in largely non-overlapping subsets of papers.

9.5.3. Safety Evaluation Coverage

Table 16 reveals what we term the *safety evaluation paradox*: 82 of 107 studies (76.6%) include no safety evaluation at all. We code a study as *dedicated* when it reports results on at least one purpose-built safety benchmark or

Table 13

Reward failure reporting ($n = 107$).

Reported	Count	%
Yes	56	52.3
No/N/A	51	47.7

Table 14Evaluation paradigm distribution ($n = 107$).

Type	Count	%
Automatic only	77	72.0
Hybrid (human + auto-matic)	26	24.3
N/A (survey)	4	3.7

Table 15Top benchmark usage frequency ($n = 107$).

Benchmark	Count	Domain
MATH (Hendrycks et al., 2021)	19	Reasoning
GSM8K (Cobbe et al., 2021)	16	Reasoning
AlpacaEval 2	14	Alignment
MT-Bench	13	Alignment
HumanEval	10	Code
Arena-Hard	9	Alignment
AIME 2024	8	Competition math
MMLU	8	Knowledge
TruthfulQA	8	Safety/factuality
Open LLM Leaderboard	7	Aggregate
Reddit TL;DR (Stiennon et al., 2020)	6	Summarization
HotpotQA	5	Agent/QA
BIG-Bench Hard	5	Reasoning
Custom red-team eval	5	Safety/red-team
JailbreakBench	3	Safety/red-team
HarmBench	3	Safety/robustness
ALFWorld	3	Agent/interactive
ProcessBench	3	PRM evaluation
GPQA Diamond	3	Science reasoning
LiveCodeBench	3	Code
Codeforces	3	Competition code
AdvBench	2	Safety/red-team
WebShop	2	Agent/interactive
miniF2F	2	Formal math
WebArena (Zhou et al., 2024)	1	Agent/web
ToolBench (Qin et al., 2023)	1	Tool-use

adversarial evaluation, *partial* when safety is assessed only incidentally (e.g., a single toxicity or truthfulness metric within a broader suite), and *none* otherwise; the study-level coding is released with the evidence table (Section 1.3). All 18 safety-domain papers include dedicated safety evaluation. Among the 89 non-safety papers, only three, Llama 2-Chat (Touvron et al., 2023), Sparrow (Glaese et al., 2022), and the o1 System Card (OpenAI, 2024), include dedicated safety evaluation, four more include partial checks, and the remaining 82 (92.1%) report none. The o1 System Card is the sole reasoning-domain exception, introducing deliberative alignment for safety reasoning. This gap is especially concerning for tool-use agents (Qin et al., 2023; G. Wang et al., 2023; Yao et al., 2023) that have direct real-world interaction capabilities, where unsafe tool calls could have tangible consequences. The paradox implies that the field treats safety as a domain-specific concern rather than a cross-cutting evaluation requirement, creating a systematic blind spot for models deployed across multiple functional contexts.

9.6. Failure Modes and Mitigations

9.6.1. Failure Mode Prevalence

The distribution of failure modes is reported in Table 17. Reward hacking is the most prevalent failure mode at 28.0%, consistent with the overoptimisation scaling laws established by L. Gao, Schulman, and Hilton (2023), which demonstrate that proxy reward quality degrades predictably as KL divergence from the reference policy increases. Distribution shift and out-of-distribution bias (21.5%) constitute the second most common failure, reflecting the fundamental challenge of RL policies encountering states unseen during training. DPO-specific failures (8.4%) include length exploitation (Park et al., 2024) and OOD overfitting (Azar et al., 2024), both well-documented pathologies of offline preference methods. A novel category, “reasoning risks”, emerges from the o1 System Card (OpenAI, 2024), where advanced reasoning capabilities *increase* the model’s ability to circumvent safety measures, representing an under-explored failure mode where capability improvement leads directly to safety degradation.

9.6.2. Mitigation Strategies

Across the 107 studies, 93 mitigation *instances*, occurrences of a mitigation being applied or proposed in a

Table 16Safety evaluation coverage ($n = 107$).

Safety Eval	Count	%
Yes (dedicated)	21	19.6
Partial	4	3.7
No	82	76.6

Table 17Failure mode prevalence ($n = 107$). Studies may report multiple failure modes, so counts sum to more than the number of reporting studies; rates denote the share of the 107 studies reporting each mode.

Failure Mode	Count	Rate (%)
Reward hacking / over-optimisation (L. Gao, Schulman, & Hilton, 2023)	30	28.0
Distribution shift / OOD bias	23	21.5
Mode collapse / verbosity	11	10.3
DPO-specific failures (Azar et al., 2024; Park et al., 2024)	9	8.4
Safety regression	7	6.5
AI feedback quality degradation	5	4.7
ORM masks flawed reasoning (Uesato et al., 2022)	3	2.8
PRM annotation scalability	3	2.8
Alignment tax (Lin et al., 2024)	3	2.8
Reasoning risks (capability $\rightarrow$ risk) (OpenAI, 2024)	1	0.9

study, rather than 93 unique techniques, were identified and consolidated into nine strategy categories. The complete failure-mode-mitigation-effectiveness mapping is visualised in Fig. 10, and the underlying edge list (study, failure mode, mitigation, reported effectiveness) is released with the evidence table. In the figure, the ten fine-grained failure modes of Table 17 are consolidated into six groups by merging the four lowest-prevalence modes into their parent categories. The most common mitigation categories, counted by the number of distinct studies applying them, include KL regularisation variants (13 studies) (Bai, Jones, et al., 2022; Ouyang et al., 2022; Touvron et al., 2023), iterative/online training (13) (Calandriello et al., 2024; Dong et al., 2024), data quality curation (9) (Cui et al., 2024; Lambert et al., 2025), process supervision (6) (Lightman et al., 2024; Setlur et al., 2024; P. Wang et al., 2024), and tool augmentation for grounding (6) (Gou et al., 2024; Schick et al., 2023; Yao et al., 2023); several studies contribute multiple distinct techniques within a category, which is why the 56 study-category applications yield 93 instances in total. Three novel mitigations entered with the most recent studies: adaptive victim fine-tuning for diverse red-teaming (Yun et al., 2025), self-play against previous iterations (Z. Chen et al., 2024), and online parameter interpolation between RL policy and SFT model (Lu et al., 2024).

9.7. Reproducibility and Reporting Practices

9.7.1. Reproducibility Artefact Availability

Table 18 provides a comprehensive breakdown of reproducibility artefact availability across eight categories. Reward definition transparency (72.0%) is the only metric above 50%, driven by the theoretical explicitness of reward-free methods (Ethayarajh et al., 2024; Rafailov et al., 2023) and process reward papers (Lightman et al., 2024), which require formal specification of the reward signal as part of the

method description. At the opposite extreme, random seed reporting (5.6%) and compute budget disclosure (5.6%) are critically low, representing the most severe reporting gaps in the corpus. Code release (43.0%) and model weight release (19.6%) show moderate availability. Exemplary open-source contributions include Tulu 3 (Lambert et al., 2025), DAPO (ByteDance Seed and Tsinghua AIR, 2025), and the RLHF Workflow (Dong et al., 2024), which release code, weights, data, and hyperparameters as a unified reproducibility package. Hyperparameter completeness is notably bifurcated: only 24 papers (22.4%) provide full hyperparameter specifications, while 71 (66.4%) provide partial details, typically learning rates without batch sizes, or KL coefficients without clipping parameters.

9.7.2. Venue Tier and Reproducibility

Table 19 cross-tabulates code and model weight release rates by venue tier. Descriptively, Tier 1 conferences show the highest code release rate in this corpus at 47.8%, Tier 2 venues (TMLR, ACL Findings, COLM) follow at 42.9%, and preprints show the lowest rate at 38.9%. These figures are consistent with peer review acting as a reproducibility incentive, but they must be interpreted with caution: the strata are small (Tier 2 contains only seven papers) and purposively sampled, and venue tier is confounded with publication year, institution type, model openness, and whether a preprint later appeared at a venue. We therefore present these rates as descriptive of the corpus only and refrain from causal interpretation; a prespecified regression with uncertainty intervals over a systematically sampled corpus would be required to support stronger claims. For model weight release, Tier 2 leads by a wide margin (57.1%), while Tier 1 (13.0%) and preprints (20.4%) trail, reflecting that weight release depends more on institutional policy (open-weight vs. proprietary) than on publication venue.

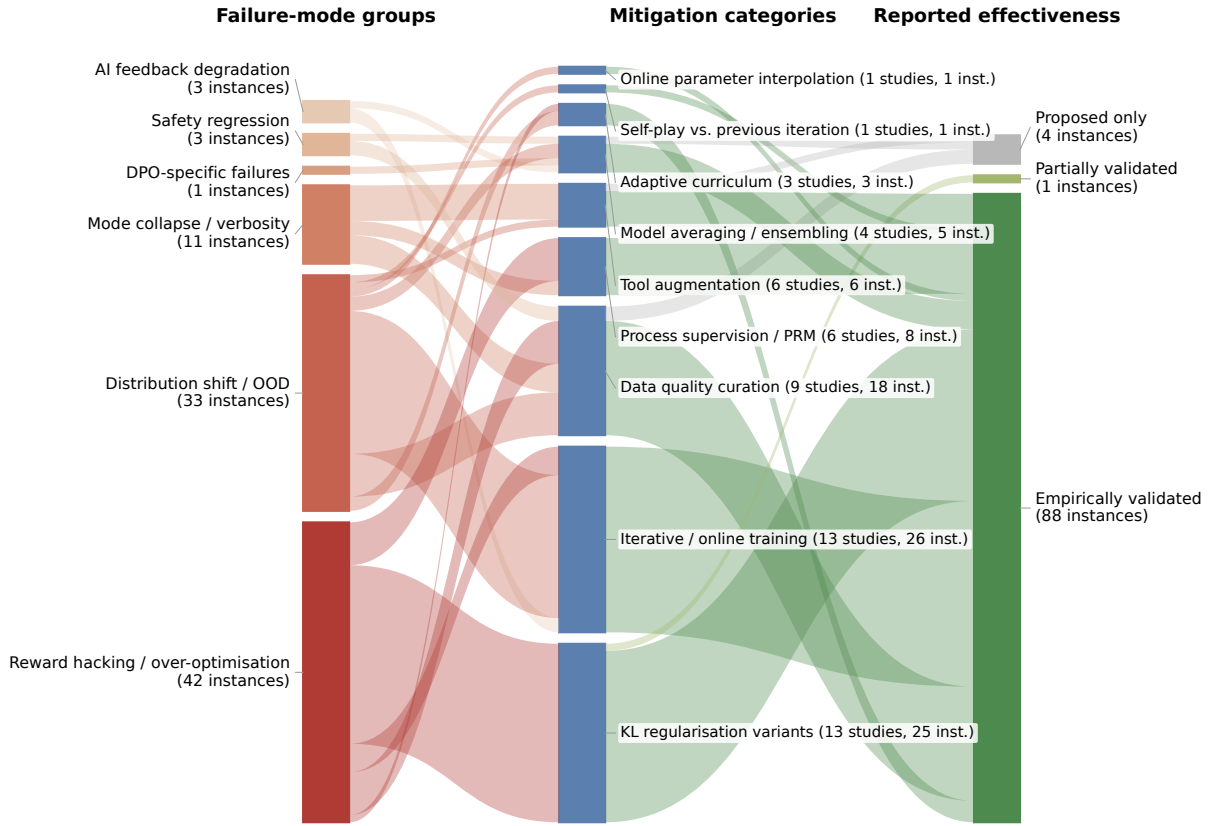


Figure 10: Failure mode → mitigation strategy → reported-effectiveness flow, generated programmatically from the released instance list (93 mitigation instances). Left: six consolidated failure-mode groups (the four lowest-prevalence modes of Table 17 are merged into their parent categories), sized by the number of instances targeting each group. Centre: the nine mitigation-strategy categories; each node is annotated with both the number of *distinct studies* applying the category and the number of *instances* it contributes (e.g., KL regularisation: 13 studies, 25 instances; iterative/online training: 13 studies, 26 instances), so node labels and ribbon widths use explicitly distinct units. Right: reported-effectiveness levels. Ribbon widths are proportional to instance counts throughout. The flow shows that most mitigation effort targets reward hacking and distribution shift, while safety regression and AI-feedback degradation receive comparatively less attention.

Table 18

Reproducibility artefact availability ($n = 107$).

Artefact	Yes	Partial	No / NR	Release Rate (%)
Code released	46	0	61	43.0
Model weights released	21	1	85	19.6
Training data public	15	12	80	14.0
Reward model data public	9	4	94	8.4
Eval prompts public	34	36	37	31.8
Hyperparameters complete	24	71	12	22.4
Reward definition transparent	77	3	27	72.0
Random seeds / runs reported	6	0	101	5.6

9.8. Five Macro-Level Patterns

The preceding analyses converge on five macro-level patterns, summarised visually in Fig. 11.

9.8.1. Pattern 1: Algorithmic Diversification and the Decline of PPO Hegemony

The most striking temporal trend in the corpus is the decline of PPO from near-universal dominance to one option among many. In the earliest studies (the 12-study 2020–2022

subset), PPO-style actor–critic optimisation, the approach that powered InstructGPT (Ouyang et al., 2022) and subsequent RLHF pipelines (Stiennon et al., 2020; Ziegler et al., 2019), was used in 9 of the 11 RL-trained systems (82%); measured in the same unit, PPO-style optimisation is used by only 35 of the 82 trained systems corpus-wide (42.7%), and PPO as a primary family accounts for 18.7% of the corpus. Given the small early-period denominator, we report this as

Table 19Code and weight release by venue tier ($n = 107$).

Tier	n	Code	Wt.	Code %	Wt. %
Tier 1	46	22	6	47.8	13.0
Tier 2	7	3	4	42.9	57.1
Preprint	54	21	11	38.9	20.4
Total	107	46	21	43.0	19.6

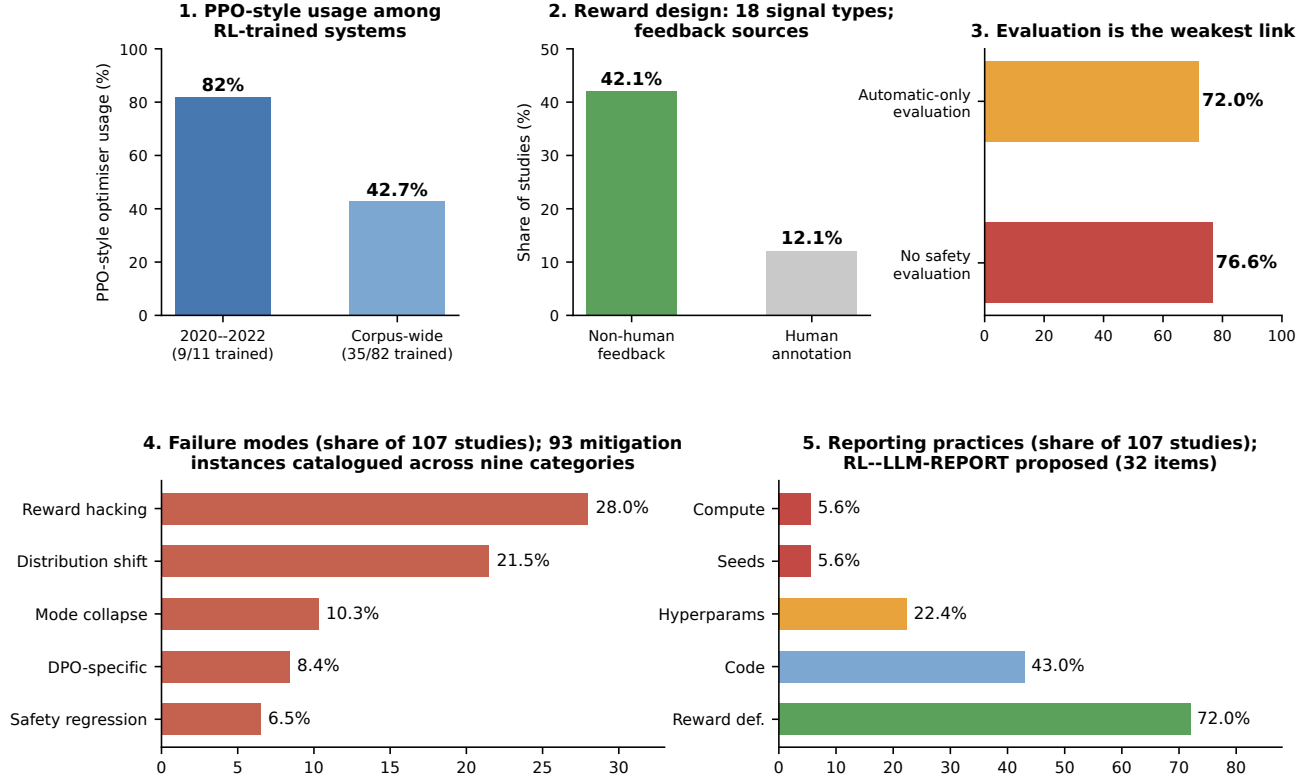
Five macro-level patterns of the cross-cutting analysis ($n = 107$ curated studies)

Figure 11: Visual summary of the five macro-level patterns ($n = 107$ curated studies), generated programmatically from the released evidence table. Panel 1: PPO-style optimiser usage among RL-trained systems falls from 82% (9/11, 2020–2022) to 42.7% (35/82) corpus-wide. Panel 2: reward design spans 18 signal types, and non-human feedback sources (42.1% of studies) far exceed human annotation (12.1%). Panel 3: 76.6% of studies omit safety evaluation and 72.0% evaluate with automatic metrics only. Panel 4: reward hacking (28.0%) and distribution shift (21.5%) lead the failure modes, countered by 93 catalogued mitigation instances. Panel 5: reporting gaps, with compute budgets (5.6%), random seeds (5.6%), and complete hyperparameters (22.4%) at critically low rates.

a qualitative trend rather than a precise estimate. This decline is not attributable to any single replacement algorithm, but rather to a *diversification wave* spanning three distinct fronts: (1) reward-free offline methods (DPO (Rafailov et al., 2023), IPO (Azar et al., 2024), KTO (Ethayarajh et al., 2024), SimPO (Meng et al., 2024), ORPO (J. Hong et al., 2024), CPO (H. Xu et al., 2024); collectively 14.0%), (2) lightweight on-policy alternatives (GRPO (Shao et al., 2024), DAPO (ByteDance Seed and Tsinghua AIR, 2025)) that eliminate the value network, and (3) domain-specific algorithms such as RLPAF for formal theorem proving (Xin et al., 2024) and classifier-based RL for red-teaming (Z.-W. Hong et al., 2024; Perez et al., 2022). The practical

implication is clear: the field has moved beyond the “one algorithm fits all” paradigm. Researchers now select algorithms based on domain-specific requirements, GRPO for reasoning (DeepSeek-AI, 2025; Shao et al., 2024), where verifiable rewards eliminate the need for learned reward models; DPO for alignment (Rafailov et al., 2023; Tunstall et al., 2023), where simplicity and stability are paramount; and PPO for safety (Dai et al., 2023; Ge et al., 2024), where online interaction with the environment remains necessary for adversarial training.

9.8.2. Pattern 2: Reward Modelling Has Become Domain-Specific

Our taxonomy identifies 18 distinct reward signal types, the largest reward-type taxonomy compiled in the RL-for-LLM literature to date. This proliferation reflects a fundamental shift: reward design is no longer a generic component but a domain-specific engineering choice. The alignment domain relies primarily on implicit DPO-style rewards and learned Bradley–Terry models (Bai, Jones, et al., 2022; Ouyang et al., 2022). The reasoning domain is bifurcating between outcome-based verification (ByteDance Seed and Tsinghua AIR, 2025; DeepSeek-AI, 2025) and step-level process supervision (Khalifa et al., 2025; Lightman et al., 2024; Setlur et al., 2024; P. Wang et al., 2024). The safety domain has developed classifier-based rewards (Z.-W. Hong et al., 2024; Perez et al., 2022; Samvelyan et al., 2024) optimised for detecting harmful content. The tool-use domain uniquely leverages environment feedback from API execution, code compilation, and game environments (Schick et al., 2023; G. Wang et al., 2023; Yao et al., 2023).

9.8.3. Pattern 3: Evaluation Is the Weakest Methodological Link

The evaluation landscape reveals three structural deficiencies. First, *benchmark fragmentation*: while MATH (19 papers) and AlpacaEval 2 (14) lead, the corpus spans over 50 distinct benchmarks with no standardised suite per domain. Second, *statistical weakness*: roughly half of the papers report only single-run benchmark scores without confidence intervals or significance tests, and a mere 5.6% report random seeds or multiple runs. Third, *safety evaluation absence*: 76.6% of papers include no safety evaluation, a finding consistent with concerns raised by Bai, Kadavath, et al. (2022) and OpenAI (2024) about the need for systematic safety assessment.

9.8.4. Pattern 4: The Safety Evaluation Paradox

A striking pattern emerges from the cross-domain analysis: safety is assessed almost exclusively by papers that target safety as their primary objective. Among the 89 non-safety-domain papers, only three include dedicated safety evaluation and four include partial checks (Table 16). This creates a systematic blind spot: most alignment methods (Meng et al., 2024; Ouyang et al., 2022; Rafailov et al., 2023) are evaluated for helpfulness but not harmlessness; reasoning models (DeepSeek-AI, 2025; Kimi Team, 2025; Shao et al., 2024) are evaluated for accuracy but not for safe reasoning steps; tool-use agents (Qin et al., 2023; G. Wang et al., 2023; Yao et al., 2023) are evaluated for task success but not for harmful tool calls. The o1 System Card (OpenAI, 2024) is the sole reasoning-domain exception, introducing “deliberative alignment” where the model reasons about safety policies before responding.

9.8.5. Pattern 5: Reporting Quality Differs Markedly Across Venue Tiers

Descriptively, code release differs across venue tiers in this corpus: Tier 1 conferences (47.8%) and Tier 2 outlets

(42.9%) show higher release rates than preprints (38.9%). These are unadjusted proportions over small, purposively sampled strata (Tier 2 comprises only seven papers), and venue is confounded with publication year, institution type, model openness, and publication status, so we draw no causal or even correlational inference from them. However, the most severe gaps are not code-related but methodological: only 5.6% of papers report compute budgets, 5.6% report random seeds, and 22.4% provide complete hyperparameters. Open-source contributions such as Tulu 3 (Lambert et al., 2025), DAPO (ByteDance Seed and Tsinghua AIR, 2025), and the RLHF Workflow (Dong et al., 2024) demonstrate that full transparency is achievable, while proprietary system reports such as GPT-4 (OpenAI, 2023) and o1 (OpenAI, 2024) contribute minimally to scientific reproducibility despite their outsized influence.

10. Practitioner Guidance

This section distills the evidence of Sections 4–9 into two actionable instruments: a decision-support matrix mapping each functional domain to recommended configurations (Section 10.1), and RL-LLM-REPORT, a proposed reporting standard for RL-based LLM research (Section 10.2).

10.1. Decision-Support Matrix

Table 20 maps each functional domain to the algorithms, reward types, KL strategies, and evaluation suites most strongly supported by the surveyed literature, together with the pitfalls most frequently reported in that domain. Practitioners should treat the matrix as a starting point rather than a prescription: as Sections 4.1 and 4.3 document, the PPO-versus-DPO trade-off depends on preference data quality and reward landscape complexity (S. Xu et al., 2024), and RainbowPO (Zhao et al., 2025) shows that tuned combinations of design components outperform any single named method. The entries also vary in evidential strength: algorithm and reward-type recommendations are supported by multiple empirical training studies per domain, whereas the KL ranges and evaluation-suite choices reflect common practice and author judgement rather than controlled comparisons; hyperparameters such as β are strongly task-, data-, and model-dependent; and the matrix does not encode compute cost or applicability conditions, which practitioners should assess against their own constraints.

To make the evidential status of each recommendation explicit, we state the eligible denominator and selection rule: for each domain, the algorithm entry is derived from the family distribution of that domain’s *trained systems* (Table 8, restricted to the domain), grouping optimiser-compatible subfamilies with their parent (e.g., PPO-based agentic RL with PPO). Under this rule the per-domain evidence is: Alignment, 36 trained systems, of which 15 use DPO-family objectives and 12 PPO; Reasoning, 18 trained systems, of which 6 use GRPO/DAPO and 4 PPO-based training; Tool-Use, 9 trained systems, of which 6 use PPO-style and 3 GRPO-style optimisation with environment

Table 20

Decision-support matrix: recommended RL configurations by functional domain, derived from the 107-study corpus. Entries vary in evidential strength (see accompanying text); they are starting points, not prescriptions.

Domain	Algorithm	Reward Type	KL Strategy	Evaluation	Known Pitfalls
Alignment	DPO/SimPO (Meng et al., 2024; Rafailov et al., 2023); PPO (Ouyang et al., 2022)	Implicit (DPO) or learned BT	β task/model-dependent (commonly 0.01–0.5); tune per setup	AlpacaEval 2, MT-Bench, Arena-Hard + TruthfulQA	Length exploitation (Park et al., 2024); reward hacking (L. Gao, Schulman, & Hilton, 2023); alignment tax (Lin et al., 2024)
Reasoning	GRPO/DAPO (ByteDance Seed and Tsinghua AIR, 2025; Shao et al., 2024); PPO+PRM (Lightman et al., 2024)	Rule-based verifiable; PRM	KL to SFT; no value net (GRPO)	MATH, GSM8K, AIME, ProcessBench	Format hacking (DeepSeek-AI, 2025); ORM masks reasoning (Uesato et al., 2022)
Tool-Use	PPO with env. reward; GRPO (J. Feng et al., 2025; Jin et al., 2025)	Environment feedback; execution success	Conservative KL	WebArena (Zhou et al., 2024), ToolBench (Qin et al., 2023)	API misuse; hallucinated tool calls; near-zero safety eval
Safety	PPO with classifier (Dai et al., 2023; Perez et al., 2022)	Classifier-based; adversarial	CMDP constraints	HarmBench, JailbreakBench	Mode collapse (Z.-W. Hong et al., 2024; Samvelyan et al., 2024)
Continual	Online DPO; iterative RLHF (Dong et al., 2024)	Progressive ref.; self-play (Z. Chen et al., 2024)	Stepwise updating	Held-out eval; capability retention	Forgetting (Qi et al., 2024); reward staleness

rewards; Continual, 4 trained systems, of which 3 use iterative or online preference optimisation. Safety has no single modal family among its 14 trained systems: PPO-based constrained training (2 studies) and classifier-reward re-teaming (2) are the most repeated patterns, and the remainder are single-study families, so the Safety algorithm entry, like the KL-practice and evaluation columns throughout, should be read as author-synthesised guidance informed by the cited works rather than a corpus mode. Reward-design entries are the modal reward types of Table 11 restricted to each domain. Beyond configuration selection, three operational practices are strongly supported by the corpus. First, *reward hacking monitoring* should be standard: ensemble reward models and tracking of proxy–gold reward divergence via the scaling laws of L. Gao, Schulman, and Hilton (2023) provide early warning of overoptimisation. Second, every RL-trained system, regardless of its primary domain, should ship with a minimal safety evaluation defined by outcomes rather than by an arbitrary resource share: at least one adversarial-robustness benchmark (e.g., a HarmBench or JailbreakBench subset), at least one refusal and over-refusal check, and explicit disclosure of the threat model and pass/fail criteria, with adversarial probing (Perez et al., 2022; Zou et al., 2023) where affordances warrant it. Third, multiple runs (at least three seeds with dispersion statistics) should be reported, following the example of DAPO (ByteDance Seed and Tsinghua AIR, 2025).

10.2. The RL–LLM-REPORT Standard

Motivated by the reporting gaps documented in Section 9.7, we propose **RL–LLM-REPORT**, a 32-item reporting checklist organised into five categories. Items are prioritised as Mandatory (M), Recommended (R), or Optional (O) based on the severity of the corresponding gap and the cost of compliance. For each item, Table 21 lists its observed prevalence across the 107-study corpus, providing

an empirical baseline against which future improvement can be measured.

An aggregate RL–LLM-REPORT score can be computed as:

$$\text{Score} = \frac{\sum_{i \in \mathcal{A}} w_i \cdot c_i}{\sum_{i \in \mathcal{A}} w_i} \times 100\% \quad (5)$$

where $\mathcal{A} \subseteq \{1, \dots, 32\}$ is the set of items *applicable* to the method under evaluation, $w_i \in \{3, 2, 1\}$ for Mandatory, Recommended, and Optional items respectively, and $c_i \in \{1, 0.5, 0\}$ for full, partial, or no coverage. The applicability rule prevents unfair penalisation: items that do not apply to a given method, for example, reward-model architecture (item 2) for implicit-reward methods, value-model specification (item 18) for value-network-free algorithms, or the annotation items (19–24) for systems trained without human annotation, are marked Not Applicable and excluded from both numerator and denominator. The weights, the partial-credit value of 0.5, and the priority labels are author-proposed conventions rather than empirically optimised choices; users who employ the score for gatekeeping should accompany it with a sensitivity analysis over these design choices. Scored retrospectively under this applicability rule across the 82 trained systems (surveys, prompting frameworks, benchmarks, and datasets are not scoreable), the median is 47.2% (range: 32.8%–85.9%). We recommend that journals and conferences adopt a minimum threshold of **70% on Mandatory items** as a submission expectation. At current rates, only 13 of the 82 scoreable trained systems (15.9%) would meet this threshold; the 25 non-applicable studies are excluded from the denominator rather than treated as failures. The systems meeting it include, notably Tulu 3 (Lambert et al., 2025), DAPO (ByteDance Seed and Tsinghua AIR, 2025), and the RLHF Workflow (Dong et al., 2024). Three items with near-zero prevalence,

Table 21

RL-LLM-REPORT: 32-item reporting standard for RL-based LLM research. Priority: M = Mandatory, R = Recommended, O = Optional. Observed prevalence: share of applicable studies fully meeting the item (resource and artefact items 17, 25, and 30–32 are computed over the full corpus of 107 studies; other items over the studies to which they apply, as recorded in the released scoring matrix).

#	Category / Item	Priority	Verification	Prevalence (%)
A. Reward Definition (6 items)				
1	Reward type (learned/implicit/rule-based/etc.)	M	Text check	97.6
2	Reward model architecture and scale	M	Text check	33.8
3	Reward training data source and size	M	Text check	32.3
4	Reward objective / loss function	M	Equation check	100.0
5	Feedback source (human/AI/rule-based)	M	Text check	86.6
6	Reward failure modes observed	R	Text check	67.1
B. Dataset Disclosure (5 items)				
7	Training data composition and size	M	Text check	12.2
8	Preference data collection protocol	M	Text check	50.0
9	Evaluation data specification	M	Text check	100.0
10	Data contamination / overlap analysis	R	Text check	9.8
11	Public data release or access instructions	R	URL check	12.2
C. KL & Optimization (7 items)				
12	RL algorithm name and variant	M	Text check	100.0
13	KL regularisation type and coefficient	M	Equation check	9.8
14	Learning rate and schedule	M	Text check	28.0
15	Batch size and number of training steps	M	Text check	1.2
16	Clipping / entropy / normalisation settings	R	Text check	3.7
17	Compute budget (GPUxhours or FLOPs)	M	Text check	5.6
18	Value model specification (if applicable)	R	Text check	19.2
D. Annotation Process (6 items)				
19	Annotator demographics and training	R	Text check	8.3
20	Inter-annotator agreement	R	Statistic	58.3
21	Annotation quality control measures	R	Text check	33.3
22	Number of comparisons / annotations	M	Number	75.0
23	Annotation interface / prompt design	O	Screenshot	0.0
24	Ethical review / IRB approval	O	Statement	0.0
E. Statistical Validation (8 items)				
25	Multiple random seeds / runs reported	M	Number	5.6
26	Confidence intervals or standard errors	M	Statistic	19.5
27	Statistical significance tests	R	Test name	0.0
28	Effect size reporting	R	Statistic	0.0
29	Ablation / sensitivity analysis	M	Section	95.1
30	Code release	M	URL check	43.0
31	Model weight release	R	URL check	19.6
32	Evaluation reproducibility package	R	URL check	25.2

compute budget, random seeds, and KL coefficient disclosure, cost essentially nothing to include and should be the community's first targets.

Two caveats govern how RL-LLM-REPORT should be used. First, regarding *provenance*: the item list and prevalence baselines are evidence-derived (each item corresponds to a reporting gap observed in the corpus), whereas the M/R/O priorities, weights, and the 70% threshold are author-proposed. The checklist has not yet undergone formal instrument development, no consensus or Delphi process with external experts, pilot testing on authors, item-level agreement measurement, or usability assessment has been performed, and it has not been formally benchmarked against adjacent frameworks such as the NeurIPS paper checklist, dataset and model documentation standards, or health-domain reporting guidelines (e.g., CONSORT-AI). We therefore present RL-LLM-REPORT as a concrete starting proposal for community refinement rather than a validated standard, and

we explicitly invite venues and working groups to subject it to such a process. Second, regarding *verifiability*: to make the retrospective scoring auditable, we release the full item-level scoring matrix (32 items $\times$ 107 studies) together with the deterministic generator that produces it (`generate_scoring_matrix.py`): every item value is a pure function of fields in the released evidence table, the item-specific mapping rules are implemented in that script and documented in the codebook, and `reproduce_tables.py` verifies on every run that the regenerated matrix is identical to the released file and that all 32 prevalence values match [Table 21](#). Source page/quote locators for each field are a direction for future releases.

11. Open Challenges and Future Directions

The cross-domain perspective of this survey exposes structural gaps that single-domain surveys cannot see. We highlight eight.

(1) Reward-free methods for reasoning. The taxonomy cross-tabulation (Table 7) shows *zero* DPO-family applications among the reasoning studies in our corpus, and early adaptations in the broader literature, Step-DPO (Lai et al., 2024) and Full-Step-DPO (H. Xu et al., 2025), which construct step-wise preference pairs for long-chain mathematics, remain preliminary. The open questions are therefore ones of maturation rather than existence: whether offline preference objectives can match on-policy RLVR at scale, whether they generalise beyond mathematics, and whether they can be systematically married to process-level supervision (Lightman et al., 2024), for example, step-wise preference pairs scored by PAVs (Setlur et al., 2024), with substantial compute-saving potential.

(2) Universal safety evaluation. The safety evaluation paradox (Section 9.5) demands lightweight, domain-agnostic safety benchmarks that any RL-for-LLM paper can adopt at negligible cost, following the deliberative-alignment template of OpenAI (2024). Standardising a minimal cross-domain safety suite (e.g., TruthfulQA + Harm-Bench subset) would close the field’s largest systematic blind spot, particularly urgent for tool-using agents with real-world affordances.

(3) Multimodal RL. Kimi k1.5 (Kimi Team, 2025) is the corpus’s only multimodal reasoning-RL system. Extending verifiable-reward RL to vision–language, audio, and embodied settings, where verification itself is harder to automate, remains largely unexplored.

(4) Multi-agent RL for LLM systems. No paper in the corpus trains *interacting* LLM agents with RL, although a nascent literature has emerged outside the corpus: ReMA trains hierarchical meta-thinking and reasoning agents with multi-agent RL (Wan et al., 2025), MARFT formalises multi-agent reinforcement fine-tuning for LLM-based agent systems (Liao, Wen, Wang, & Zhang, 2025), and cooperative MARL has been applied to LLM collaboration (S. Liu, Liang, Lyu, & Amato, 2025). The open problems are therefore not the existence of such systems but their rigour: principled credit assignment across agents, training stability, emergent-communication analysis, system-level reward design, safety of interacting agents, and external validation of claimed gains.

(5) Scalable process supervision beyond mathematics. PRMs are concentrated in mathematical reasoning, where step decomposition is natural. Transferring data-efficient generative verification (Khalifa et al., 2025; Mahan et al., 2024) to open-ended domains, law, medicine, scientific writing, requires new definitions of a “step” and of “progress” (Setlur et al., 2024).

(6) Attacker–defender equilibria. The co-evolutionary loops of Section 7, red-team, harden, repeat (Ge et al., 2024; Yun et al., 2025), lack theoretical characterisation. Whether these dynamics converge, cycle, or escalate, and what game-theoretic guarantees are achievable against optimisation-based attacks (Zou et al., 2023), are unanswered questions with direct deployment consequences.

(7) Continual and lifelong post-training. The continual-improvement domain (Section 8) remains embryonic: reward staleness under distribution drift, capability retention across update cycles (Lu et al., 2024; Qi et al., 2024), and *lifelong safety*, whether hard-won robustness survives subsequent updates, have almost no longitudinal evidence. Tracking whether reward hacking (L. Gao, Schulman, & Hilton, 2023) and alignment tax (Lin et al., 2024) accumulate over deployed model generations is an essential empirical programme.

(8) Methodological consolidation. Finally, the reporting crisis (Section 9.7) is itself a research obstacle: without compute budgets, seeds, and complete hyperparameters, algorithmic comparisons across the literature are confounded. Community adoption of RL–LLM-REPORT, or any comparable standard, together with standardised per-domain benchmark suites, is a precondition for cumulative science in this field.

12. Conclusion

This survey provided a cross-domain synthesis of reinforcement learning for large language models that couples a unified taxonomy with a quantitative methodological audit and a reporting standard, a combination that, per the dated update search of Section 1.3, we have not found in any single prior survey, organising a curated corpus of 107 representative studies (2017–2026) into a unified taxonomy of five functional domains, alignment, reasoning, tool use and agentic behaviour, safety and robustness, and continual improvement, and examining each in depth. Across domains, a coherent picture emerges. Algorithmically, the field has moved from early PPO hegemony (PPO-style optimisation in 9 of the 11 RL-trained systems of the small 2020–2022 subset, against 35 of 82 trained systems corpus-wide) to a diversified landscape in which DPO-family methods dominate offline alignment, GRPO/DAPO dominate verifiable-reward reasoning, and classifier- and constraint-based PPO variants persist in safety. Reward design has matured into a domain-specific engineering discipline spanning 18 signal types, with non-human feedback now substantially exceeding human annotation. Evaluation, by contrast, has not kept pace: benchmarks are fragmented, statistical practice is weak, and safety assessment is confined almost entirely to papers whose primary topic is safety. Reporting practices lag furthest of all, with compute budgets and random seeds disclosed by barely one paper in twenty.

Beyond the synthesis itself, the survey contributes a dual-axis taxonomy, an 18-type reward-signal taxonomy, a practitioner decision-support matrix, and the RL–LLM-REPORT 32-item reporting standard with prevalence base-lines, together with eight open challenges spanning reward-free reasoning, universal safety evaluation, multimodal and multi-agent RL, scalable process supervision, adversarial equilibria, lifelong post-training, and methodological consolidation.

Scope and limitations. The corpus underlying this survey was assembled by expert curation rather than protocol-driven database search, and several consequences follow. *Selection and database bias:* candidate discovery relied on major-venue proceedings, arXiv, and citation snowballing, and may underrepresent peripheral venues, non-English publications, and sub-1B-parameter research. *Citation and influence bias:* prioritising widely adopted work systematically favours visible papers and, together with the deliberate balancing of domains, makes the sample purposive; all corpus statistics are therefore descriptive of this curated sample and must not be generalised to the field as population estimates or trends. *Temporal and version drift:* many corpus papers are evolving arXiv preprints; we code the version available at extraction time, and duplicate preprint/conference versions of the same work were consolidated into a single study, but residual version effects (e.g., results revised between versions) cannot be excluded. *Forced single-domain classification:* some influential papers span multiple domains (e.g., DeepSeek-R1 (DeepSeek-AI, 2025) contributes to both reasoning and alignment); we classify by primary contribution, which discards secondary contributions. *Coder subjectivity:* attribute extraction involves judgement (e.g., partial versus dedicated safety evaluation), and although coding was cross-checked between authors and is released for audit, no formal inter-rater reliability study with independent external coders was conducted. *Non-independence:* several corpus papers originate from the same research programmes (e.g., the DeepSeek and Qwen lineages), so studies are not independent observations. *Publication bias:* negative results are underrepresented in the underlying literature itself, Tulu 3 (Lambert et al., 2025) stands out for explicitly reporting methods that did not work. Finally, the field evolves at extraordinary speed, and work appearing after early 2026 is not captured.

Reinforcement learning has transformed large language models from powerful but uncontrollable text generators into aligned, reasoning, tool-using, and increasingly safe AI systems. Yet the field’s rapid progress has outpaced its methodological foundations. The taxonomy, analysis, and instruments provided by this survey are intended as tools for consolidation, enabling the field to build on solid empirical ground as it advances toward more capable and trustworthy AI systems.

Abbreviations

All abbreviations used in this manuscript are defined in Table 22.

Table 22
Abbreviations

Abbreviation	Definition
RL	Reinforcement Learning
LLM	Large Language Model
RLHF	Reinforcement Learning from Human Feedback
RLAIF	Reinforcement Learning from AI Feedback
RLVR	Reinforcement Learning with Verifiable Rewards
DPO	Direct Preference Optimization
PPO	Proximal Policy Optimization
GRPO	Group Relative Policy Optimization
A2C	Advantage Actor–Critic
RM	Reward Model
PRM	Process Reward Model
ORM	Outcome Reward Model
GenRM	Generative Reward Model
BT	Bradley–Terry
SFT	Supervised Fine-Tuning
CoT	Chain of Thought
MDP	Markov Decision Process
KL	Kullback–Leibler (divergence)
OOD	Out-of-Distribution
MoE	Mixture of Experts
AI	Artificial Intelligence
NLP	Natural Language Processing
NR	Not Reported
N/A	Not Applicable
GSM8K	Grade School Math 8K Benchmark
MATH	Mathematics Benchmark Dataset

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